

Information and Macroeconomic Expectations: Global Evidence*

Francesco D'Acunto[†] and Michael Weber[‡]

This version: August 2026

Abstract

Unique global data on 47,000 consumers across 47 countries representing 90% of world GDP reveal universal patterns in macroeconomic belief formation. Most consumers form beliefs based on local information sources (utility bills, shopping, and acquaintances) that provide signals not representative for the overall consumption bundle, so that seeking information leads to forming biased macroeconomic expectations. Distrust in governments and central banks, the producers of aggregate economic information, helps explain reliance on local sources and the lack of learning from aggregate realizations. Observed macroeconomic volatility fuels distrust. In the cross-section, we confirm that certain demographics correlate with expectation biases but mostly due to different sorting into information sources, rather than different processing of the same economic information. These globally established facts inform advances of theories of belief formation and the design of communication strategies as a macroeconomic policy tool.

JEL classification: C90, D14, D84, E31, E52, E71, G11

Keywords: Behavioral Macroeconomics, Behavioral Finance, Subjective Beliefs, Traditional Media, Social Media, Rational Inattention, Trust, Information Economics, Institutions, Surveys.

*We thank Tommaso Manfè, Gokul Raj Boobathy, and Carlos Chavez for excellent research assistance. For very helpful comments and discussions, we thank Winnie Coleman, Fiorella de Fiore, Yuriy Gorodnichenko, and conference and seminar participants in the 2025 American Economic Association Annual Meeting, the G53 Financial Literacy seminar, the Virtual Asian Seminar on Macroeconomic Expectations, the Inter-American Development Bank, Texas A&M, Verein fuer Socialpolitik, European Central Bank and Federal Reserve Bank of Cleveland Inflation Conference, BIS Asian Expectation Network, Econometric Society World Congress, Society of Economic Dynamics Conference, CESifo Area Conference on Macro with Micro Data 2026, KOF Institut ETH Zurich, Barcelona Summer Forum, Expectations in Dynamic Macroeconomic Models Conference, Federal Reserve Bank of Cleveland, SAFE Household Finance Conference, 2nd Kiel-CEPR Conference on Monetary Policy, and the University of Macau. We thank the Alfred P. Sloan Foundation for the generous funding under grant #G-2022-19504, the National Science Foundation under grant #2149129, and Morning Consult for data collection. Weber also gratefully acknowledges the funding by the Alexander von Humboldt Foundation and the German Federal Ministry of Research, Technology and Space.

[†]McDonough School of Business, Georgetown University, and CEPR. e-Mail: francesco.dacunto@georgetown.edu.

[‡]Daniels School of Business, Purdue University, ESMT Berlin, CEPR, and NBER. e-Mail: weber366@purdue.edu.

“I think the people in [the UK] have had enough of experts, from organisations with acronyms, saying that they know what is best and getting it consistently wrong.”

Michael Gove, UK Secretary of State for Justice, June 2016

“China and its ruling clique, the Chinese Communist Party, are continuing a long tradition of communist or socialist authoritarian nations faking economic statistics.”

Keith Naughton, The Hill, May 2025

“In my opinion, today’s Jobs Numbers were RIGGED [...] – A TOTAL SCAM”

Donald J. Trump, Truth Social, August 1, 2025

1 Introduction

Most models in macroeconomics and finance assume that agents form beliefs using unbiased information about aggregate economic fundamentals, even when relaxing the assumption of full-information rational expectations (FIRE).¹ Empirical studies estimate the costs and drivers of accessing economic information sources (Fuster et al., 2022; Faia et al., 2024). Once accessed, sources are typically assumed to provide possibly noisy but unbiased signals about the aggregate variables consumers forecast. An important benchmark is the epidemiological model of Carroll (2003), in which households periodically update their expectations based on professional forecasts transmitted through the news media. This framework allows households to remain inattentive and generates sluggish adjustment in aggregate expectations, but the information transmitted through the media remains an aggregate signal.

This abstraction is not innocuous in a world in which households often do not form beliefs based on aggregate statistics or central-bank communication (Lamla and Vinogradov, 2019;

¹In sticky-information, dispersed-information, and rational-inattention models, agents may update infrequently, observe noisy signals, or optimally allocate limited attention across state variables (Sims, 2003; Woodford, 2003; Reis, 2006; Mankiw et al., 2004; Maćkowiak and Wiederholt, 2009; Paciello and Wiederholt, 2014; Afrouzi and Yang, 2021; Maćkowiak et al., 2023). In models of higher order beliefs and with imperfect common knowledge, agents may be uncertain about others’ information or responses (Morris and Shin, 2002; Angeletos and Pavan, 2007; Nimark, 2008; Angeletos and Lian, 2018; Angeletos and Huo, 2021). In behavioral macroeconomic models, agents may overreact, underreact, extrapolate, or rely on subjective mental representations of the economy (Gennaioli and Shleifer, 2018; Bordalo et al., 2022; Bianchi et al., 2024; L’Huillier et al., 2024; Angeletos et al., 2021; Kohlhas and Walther, 2021; Andre et al., 2022).

De Fiore et al., 2024), but rely on signals encountered in daily life, or from local sources of economic information: shopping, utility bills, social interactions, work-related activities, and social media (Coibion and Gorodnichenko, 2015; Cavallo et al., 2017; Bailey et al., 2018; Kindermann et al., 2021; D’Acunto et al., 2021; Kamdar and Ray, 2022; Wehrhöfer, 2026; Patzelt and Reis, 2024). Contrary to the standard assumption in a Lucas island model, these local sources do not necessarily provide unbiased signals of aggregate statistics. Items observed during shopping activities, for instance, are often non-representative of the aggregate or even individual consumption bundle (Anesti et al., 2024). And, they relate to bundle components that face especially volatile price changes (D’Acunto et al., 2021). If agents rely on such local signals, acquiring more information might not necessarily result in more accurate perceptions and expectations of macroeconomic variables such as inflation, interest rates, and house prices (Kuchler and Zafar, 2019; D’Acunto et al., 2021; Kuchler et al., 2023; Anesti et al., 2024).

To study if, how, and why heterogeneous information sources shape the cross-section of consumers’ subjective expectations, we designed and implemented, to the best of our knowledge for the first time, a global homogeneous survey of consumers’ macroeconomic perceptions, expectations, information sources, and trust in economic institutions. The survey covers about 47,000 respondents across 47 countries representing roughly 90% of world GDP. The global scope is essential. Existing work has documented large and persistent deviations from FIRE in individual countries, including demographic differences in inflation expectations, heterogeneous responses to information treatments, and systematic extrapolation from salient prices. A single-country design, however, cannot easily distinguish institutional, cultural, political, and macroeconomic explanations. Our setting allows us to assess which features of belief formation arise universally, even among agents facing very different macroeconomic histories, economic systems, institutional environments, and media.

We start by documenting three sets of facts. First, and almost universally, consumers’ perceptions and expectations of price-related macroeconomic variables are on average biased upwards relative to ex-post realizations. The bias does not only arise for inflation, but also for macroeconomic variables whose definitions are more similarly and easily understood by

consumers, such as interest rates and house prices. Second, the sources consumers use to form macroeconomic beliefs differ starkly both within and across countries. Some rely on aggregate sources, such as official reports, government statistics, central-bank information, newspapers, and television. Others rely on local sources, such as shopping activities, utility bills, social and work-related interactions, and social media. We show that consumers who rely on aggregate sources form more accurate perceptions of current macroeconomic variables than those who rely on local sources, all else equal. Third, the type and intensity of information-source use help to explain expectation biases: consumers who rely on local information sources form more positively biased expectations relative to ex-post realizations than those who rely on aggregate sources. And, among consumers who rely on the same local source, which keeps constant unobserved correlates of information source selection, more intensive exposure is associated with larger biases.

These motivational baseline facts speak to several classes of models of expectations formation. In standard rational-inattention models, agents choose how much attention to allocate to collecting economic information subject to cognitive or information-processing constraints. More generally, models of endogenous information acquisition allow agents to choose information about both economic fundamentals and endogenous aggregate outcomes, with the structure of information costs shaping equilibrium efficiency and nonfundamental volatility ([Hébert and La'O, 2023](#)). In dispersed-information and higher-order-belief models, agents observe heterogeneous signals and care about others' beliefs. These frameworks show that the precision of agents' information about fundamentals need not determine the dynamics of higher-order beliefs and can generate belief inertia, disagreement, and muted aggregate responses to shocks because of imperfect common knowledge ([Morris and Shin, 2002](#); [Angeletos and Pavan, 2007](#); [Nimark, 2008](#); [Angeletos and La'O, 2009](#); [Angeletos and Lian, 2018](#); [Angeletos and Huo, 2021](#)). These models can therefore explain why agents remain imperfectly informed and why information frictions have important aggregate consequences. The signals agents observe, however, are typically correctly specified, although potentially noisy, indicators of the relevant aggregate state. Our results point to an additional friction: consumers may select information sources

that provide systematically unrepresentative signals about the macroeconomic variables they seek to forecast. Acquiring more information can then increase, rather than reduce, belief biases.

These distinctions clarify the relationship between our baseline evidence and recent work on media, public information, and learning from prices. [Chahrour et al. \(2021\)](#) show that even accurate media coverage can generate aggregate fluctuations when attention is focused on newsworthy but unrepresentative sectors. [Gaballo \(2016\)](#) studies how public communication interacts with rational inattention and market-based learning. [Chahrour and Gaballo \(2021\)](#) show that agents can learn from prices in ways that amplify business-cycle fluctuations. Our paper provides direct global micro evidence on the household side of this mechanism. Consumers attend to concrete and salient local signals that may be informative about their personal environment but are not representative of the aggregate object they need to forecast to make important decisions, such as investment, housing, and debt choices, among many others. The resulting expectation errors are therefore not simply the consequence of insufficient information, but of reliance on non-representative information.

As a second step, we make use of the cross-sectional component of our data to suggest a unifying interpretation of why demographic characteristics appear to correlate systematically with biases in macroeconomic expectations. Prior work has documented systematic differences in inflation expectations by gender, income, race, education, financial literacy, and cognitive abilities ([Jonung, 1981](#); [Bryan and Venkatu, 2001](#); [Pfajfar and Santoro, 2009](#); [Bruine de Bruin et al., 2010](#); [D’Acunto et al., 2021](#); [D’Acunto et al., 2023](#); [D’Acunto et al., 2022](#); [Coibion et al., 2020](#)). Closely related to our analysis, [Conrad et al. \(2022\)](#) show for Germany that the information channels households use to learn about monetary policy are systematically related to their socioeconomic characteristics and, in turn, to their inflation perceptions, expectations, and uncertainty. We provide global evidence that many of these demographics explain sorting into different information sources. Once we condition on the sources consumers use, though, most demographics are barely related to differences in expectations. This result does not imply that demographics are irrelevant. It suggests that demographics might matter primarily

because they predict the information sources consumers use when forming beliefs rather than differences in processing the same information.

Finally, we investigate why consumers around the globe rely on local information sources despite the widespread availability of aggregate information. Standard models of endogenous information acquisition emphasize that households compare the expected benefits of information with the costs of acquiring and processing it. Consistent with this mechanism, [Roth et al. \(2022\)](#) show experimentally that households who learn that they face greater personal unemployment risk during recessions increase their demand for an expert macroeconomic forecast. Our analysis considers a distinct but complementary margin. Rather than asking whether households acquire macroeconomic information, we study which sources they use and why they may rely on local signals instead of more representative aggregate information.

Differences in conventional search costs alone seem unlikely to explain this source choice. In the information age, in which the majority of households regularly access the internet, aggregate statistics are widely and inexpensively available. Accessing aggregate inflation information can require an action as simple as googling the term “inflation” and reading/recalling a single number that appears on the screen almost instantaneously. Local information, by contrast, is often encountered passively, but collecting, recalling, and processing dozens of prices observed while shopping, paying utility bills, or interacting with acquaintances into an assessment of an aggregate macroeconomic variable requires substantial cognitive effort.

An alternative route to explain why consumers do not rely on aggregate information is that they might not believe it is a truthful representation of reality. We therefore advance trust in aggregate economic information as a potential mechanism. In each country, governments and central banks produce and disseminate key aggregate economic information. If consumers distrust these institutions, and believe that the official statistics they produce are plagued by conflicts of interest and are “rigged,” as the President of the United States recently stated (see opening quotes), they may discount or outright dismiss aggregate economic information and place greater weight on local information sources, which they can access and witness personally without any intermediation by economic institutions. Relatedly, [Cavallo et al. \(2016\)](#) show that

Argentine households exposed to potentially manipulated inflation statistics neither accepted official information at face value nor discarded it entirely, but attempted to correct for its perceived bias. We study a complementary margin: whether distrust in the producers of aggregate information predicts the sources consumers choose when forming macroeconomic beliefs. This channel could also rationalize why consumers do not appear to learn that local information sources are biased: even the ex-post aggregate realizations that reveal the bias are produced and disseminated by institutions they distrust, and hence the data are rejected.

Our evidence supports this mechanism. Trust in governments and central banks varies substantially both across and within countries. Lower trust is associated with greater reliance on local information sources and with higher perception and expectation biases. Importantly, trust in public economic institutions has explanatory power beyond trust in private economic institutions, such as banks, corporations, and the stock market (Guiso et al., 2008), and beyond whether the respondent’s preferred political party is in power. We also show that macroeconomic volatility, rather than the level of macroeconomic variables, predicts lower trust in economic institutions across countries. This pattern suggests that unstable macroeconomic environments may erode the credibility of economic institutions and the aggregate information they produce, thereby affecting the sources consumers use to form beliefs and, ultimately, the accuracy of such beliefs.

The paper contributes to three literatures. First, it contributes to the empirical literature on subjective expectations by providing homogeneous global evidence on macroeconomic perceptions, expectations, information sources, and institutional trust. Recent reviews of this literature include Armantier et al. (2013); Weber et al. (2022); D’Acunto et al. (2023); Haaland et al. (2023); D’Acunto and Weber (2024); Dräger and Lamla (2024); D’Acunto et al. (2024); Fofana et al. (2025). Existing work has largely studied expectation formation within individual countries or monetary areas.² We confirm existing facts about beliefs and provide novel global facts on the relationship between the choice of economic information sources, its drivers, and

²Dräger et al. (2026) provides a recent overview of systematic differences in inflation expectations across countries covered by the European Central Bank Consumer Expectations Survey.

its effects on expectation biases.

Second, the paper contributes to recent work in the area of behavioral macroeconomics and, more broadly, to the literature that studies tractable and plausible deviations from FIRE.³ Our evidence suggests that models should distinguish between three objects that are often collapsed into one or are so far fully absent: attention, signal precision, and signal representativeness. Standard information-friction models emphasize attention and signal precision. Our findings show that signal representativeness is also first order: a consumer can be attentive and nevertheless become more biased if attention is allocated to signals from local economic environments that are systematically unrepresentative of the aggregate variable. This mechanism connects information-friction models to theories in which selective memory or neglect of the process generating observed information produces systematic belief distortions (Gennaioli and Shleifer, 2010; Enke, 2020). Our evidence identifies information-source choice as an empirical channel through which consumers may repeatedly encounter, recall, and overweight a selected subset of signals when forming macroeconomic beliefs.

Third, the paper contributes to the literature on macroeconomic communication and policy transmission.⁴ If trust determines whether households use information from aggregate information sources, communication policy cannot be reduced to increasing the availability of official information (Christelis et al., 2020; D’Acunto et al., 2021; Ehrmann, 2025). Conditional on reaching consumers, clear explanations of a central bank’s objective and the stabilizing role of monetary policy can raise perceived credibility and improve the anchoring of inflation expectations, including among less financially literate households (Ehrmann et al., 2025). Our results highlight a complementary margin that arises before households process the content of a communication: consumers must first consider the institution and the information source sufficiently credible to rely on the message. The policy problem is therefore not only to provide clear aggregate information, but also to make that information credible and salient relative to the local signals households encounter in daily life. This perspective gives governments and

³Angeletos and Lian (2023) and D’Acunto and Weber (2026) provide recent reviews of this literature.

⁴See Blinder et al. (2024) and D’Acunto et al. (2024) for recent reviews.

central banks a role not only as providers of public information but also as institutions whose credibility shapes the information households choose to use. Restoring credibility may therefore improve the alignment of household expectations with aggregate conditions and strengthen the transmission of fiscal and monetary policy (D’Acunto et al., 2021).

2 Information from Aggregate vs. Local Sources

We follow the taxonomy proposed by D’Acunto and Weber (2024), who distinguish between two broad classes of economic information sources: aggregate and local sources. The distinction is not whether information is public or private, nor whether signals are more or less noisy. Rather, it concerns whether a source is designed to provide a representative signal about the aggregate macroeconomic variable households are asked to assess. Below, we discuss the taxonomy and its implications, while Section 4 provides empirical validation by documenting that consumers who use local information sources form directionally biased, and not just noisier, perceptions about current macroeconomic variables, whose realizations based on official statistics are public information.

Aggregate information sources provide signals that are constructed to summarize economy-wide dynamics. Examples include official statistics, government reports, central-bank forecasts and communications, and traditional media coverage that reports or interprets such aggregate information. Traditional media do not necessarily transmit aggregate information mechanically: both the intensity and tone of inflation reporting and the topics selected for coverage shape households’ inflation expectations and the degree of information rigidity (Lamla and Lein, 2014; Larsen et al., 2021). We nevertheless classify traditional media as an aggregate source because the information they report typically concerns economy-wide developments rather than signals drawn from a household’s own consumption, workplace, or social environment. These sources provide processed signals, typically summarized into one or a small set of numbers. Households incur search costs to access them and cognitive costs to understand and incorporate them into their beliefs.

By contrast, local information sources provide signals households encounter in their own economic environment. They include shopping activities, utility bills, social interactions, work-related activities, such as conversations with customers, suppliers, or employers, and salient examples leading to viral debates on social media.⁵ These signals are observed directly and are often available without deliberate search. At the same time, recalling and processing them requires substantial cognitive effort given their abundance and the lack of pre-processing embedded in aggregate statistics. For instance, for the case of shopping activities, consumers would need to recall, process, and aggregate a myriad of signals using some rule of thumb. Crucially, these signals need not be representative of the aggregate object households are asked to forecast. A household’s grocery basket, electricity bill, rent, commute, or workplace price exposure can be unrepresentative of the aggregate consumption bundle or national macroeconomic conditions (Anesti et al., 2024). Moreover, to lower the cost of processing this disaggregated information, households tend to use rules of thumb such as placing disproportionate weight on frequently observed price changes and on salient price increases relative to equal-sized price decreases. These rules of thumb produce additional systematic biases in perceptions and beliefs (D’Acunto et al., 2021; D’Acunto and Weber, 2023). This mechanism is related to theories in which agents form beliefs from a selected subset of potentially relevant information. In the local-thinking model of Gennaioli and Shleifer (2010), limited and selective retrieval from memory causes representative information to receive disproportionate weight in intuitive judgments. Similarly, Enke (2020) shows that individuals often fail to account sufficiently for the process determining which observations enter the sample they see and consequently form biased beliefs from selected information. Our setting introduces a complementary source-selection margin: consumers systematically rely on local economic signals that are readily observed or recalled but may be unrepresentative of the aggregate macroeconomic distribution they seek to assess.

Our taxonomy is also related to, but distinct from, the local-signal structure in Lucas (1975)’s islands model. In that framework, agents observe prices in their own local market

⁵Note that social media can also provide access to aggregate sources, for instance if consumers follow central banks’ accounts.

(island) and use them to infer the aggregate price level. Local prices combine an aggregate component with an island-specific component, but the island-specific component averages out across locations. Local prices are therefore noisy signals of the aggregate price level, not systematically biased indicators of the object agents aim to infer. Our taxonomy relaxes this representativeness assumption. Local sources may be not only noisy but also systematically unrepresentative of the aggregate variable’s distribution.

Inflation illustrates the distinction. A consumer who forms beliefs from grocery prices, gasoline prices, rents, or utility bills observes prices that are salient, frequent, and personally relevant. Existing evidence shows that households place substantial weight on such specific prices when forming inflation expectations (De Bruin et al., 2011; Cavallo et al., 2017; D’Acunto et al., 2021, 2022; Dietrich et al., 2022; Binder and Makridis, 2022; Weber et al., 2023). Yet these prices represent selected components of the overall consumption bundle (Anesti et al., 2024), some of which are especially volatile. Hence, local signals may be systematically tilted toward price changes that are more salient, more volatile, or more frequently observed relative to others.⁶

In standard dispersed-information and rational-inattention models, agents may observe different signals, update slowly, or allocate limited attention across state variables (Mankiw et al., 2004; Sims, 2003; Woodford, 2003; Reis, 2006; Maćkowiak and Wiederholt, 2009; Afrouzi and Yang, 2021; Paciello and Wiederholt, 2014). These models can generate disagreement, sluggish adjustment, and forecast errors. But when signals are unbiased indicators of the relevant aggregate state, acquiring more information should improve accuracy on average. If households instead obtain information from local sources that are systematically unrepresentative, greater information acquisition can increase rather than reduce bias. Consumers may be attentive, but attentive to the wrong object. In principle, households could correct for the

⁶In canonical noisy-information or signal-extraction models with correctly specified and unbiased signals, agents optimally place greater weight on signals that are observed more frequently or perceived to be more precise. Greater weight on such signals can generate disagreement, volatility, or ex-post forecast errors, but it should not generate systematic bias on average. Our mechanism differs because frequently observed local signals may be systematically unrepresentative of the aggregate macroeconomic variable households are asked to forecast.

non-representativeness of local information. Doing so would require them to understand the mapping between their local signals and the aggregate statistic, know the composition of the relevant aggregate index, remember and aggregate many local signals, and adjust for the fact that these signals are only partial indicators of the macroeconomic object of interest. This task is more complex than recalling a single aggregate statistic reported by an official source or by the media.

As discussed in the introduction, cognitive costs alone are unlikely to explain reliance on local information. Aggregate statistics are widely available, especially in countries with broad internet access and active media coverage. This is why, in our setting, we consider a different friction: households may differ in whether they view aggregate information as credible. Governments and central banks produce and disseminate much of the aggregate information used to evaluate macroeconomic conditions. Households might distrust these institutions, and think that their actions, including the production and dissemination of economic information, is determined by conflicts of interest, such as supporting unpopular policy measures or favoring economic elites over the average consumer. Official statistics would thus be considered uninformative. Local information sources, which consumers directly observe and witness, are immune to the effects of distrust. Distrust thus creates a wedge between the objective informativeness of a source and the informativeness consumers perceive.

The same logic can help explain why households may not learn that local signals are biased indicators of macroeconomic aggregates over time. In canonical learning models, agents observe realizations of the aggregate variable, compare them with prior signals, and adjust the weights they place on different sources. This process does not necessarily happen if households distrust the institutions that produce economic information, because the data is rejected. If anything, discrepancies between official statistics and personally observed economic conditions may reinforce reliance on local sources, rather than alerting households that local sources are inaccurate.

Our empirical analysis assesses three implications of this framework. First, households who rely on aggregate sources should form more accurate macroeconomic perceptions and ex-

pectations. Instead, households who rely on local sources should form more biased, not just noisier, macroeconomic perceptions and expectations. Second, households’ choice of economic information sources should help explain systematic differences in the emergence of expectation biases across demographics, above and beyond potential differences in processing the same information across demographics. Third, trust in the institutions that produce aggregate economic information—governments and central banks—should help explain households’ choice of economic information sources as well as perception and expectation biases.

3 Global Subjective Expectations Survey

We collaborated with Morning Consult, a U.S.-based market insights company that regularly conducts international surveys ([Hajdini et al., 2024](#)). We selected countries to cover a large share of global GDP, span all continents, and include economies with different inflation histories and institutional environments. The final set of countries was restricted to those for which Morning Consult could recruit sufficiently large samples and implement country-level demographic quotas. Morning Consult uses stratified sampling and works with several panel providers to maintain a stable sample composition across countries.⁷

We designed a survey to elicit homogeneously across 47 countries consumers’ subjective perceptions and expectations about macroeconomic variables, their sources of economic information, trust in institutions, and demographic and financial characteristics. The targeted average completion time was approximately 15 minutes. The institutional review board (IRB) of the University of Chicago approved the survey instrument and project under protocol number IRB22-1574. The costs of translating the survey, recruiting respondents, collecting the micro data, and preparing the data were covered by a grant from the Alfred P. Sloan Foundation.

To make the samples broadly representative of each country’s online population, sampling

⁷Morning Consult calculates weights using iterative post-stratification (raking) to targets from up-to-date population targets based on census estimates for age, gender, education, region, as well as race and ethnicity.

was stratified at the country level based on demographic characteristics. The survey used 12–16 quota cells in each country, based on two groups for sex assigned at birth, three to five age groups (e.g., 18–34, 35–44, 45+), and two educational-attainment groups defined by college completion. Age and education thresholds vary slightly across countries.

Morning Consult translated the survey into the relevant national languages and launched the first fielding on April 11, 2023. Data collection concluded on May 24, 2023. Fielding times varied across countries, reflecting differences in sample availability and the time needed to satisfy country-level quotas.

The final dataset includes consumers from 47 countries across all five continents, representing more than 75% of the world’s population and 90% of global GDP measured in U.S. dollars. For each country, the targeted sample size was 1,000 respondents, with the exception of Morocco (750 respondents) and Zimbabwe (500 respondents).⁸ The final sample consists of 46,285 observations.

3.1 Survey Structure and Questions

Given the available budget and the cross-country scope of the project, we designed the survey to target an average completion time of 15 minutes per respondent. This constraint allowed us to include 34 questions, comparable to other recent household surveys on expectations and economic decision-making (D’Acunto et al., 2021; Coibion et al., 2022).

The survey consisted of four parts.⁹ The first part elicited respondents’ demographic characteristics, including age, gender, employment status, marital status, education, field of study for college graduates, household size, and income. Income brackets were defined to match country-level income quartiles as closely as possible.

The second part elicited household financial characteristics. These included the availability of rainy-day funds, accumulated wealth relative to household income, homeownership and

⁸Final sample numbers vary slightly in some countries because multiple respondents started the survey shortly before the country-level target was reached. For instance, we obtained 1,007 responses in the U.S. and 1,002 in Spain.

⁹We report all questions in the Online Appendix.

mortgage status, and the allocation of shopping and financial decision-making responsibilities within the household.

The third part elicited numerical macroeconomic perceptions and expectations. To ensure that respondents were familiar with the concept of inflation and with the required response format in percentage changes, we began this block with the following explanation:

Question 1 *As you may know, inflation is the percentage rise in overall prices in the economy. For example, if a price increases from 10 to 11, the rate of inflation is $(11-10)/10 = 10\%$. If a price decreases from 10 to 9, the rate of inflation is $(9-10)/10 = -10\%$.*

We provided examples of both inflation and deflation to avoid priming respondents toward positive price changes (Weber et al., 2022; D’Acunto et al., 2022).

For inflation perceptions and expectations, we followed standard question wordings in the literature. In particular, we asked directly about inflation, as in the New York Fed Survey of Consumer Expectations (see, e.g., Crump et al., 2022), rather than about changes in the prices of goods respondents typically purchase, as in the Michigan Survey of Consumers. This design choice is important for our setting because asking about frequently purchased goods may prime respondents to think disproportionately about groceries and other salient prices (De Bruin et al., 2011; D’Acunto et al., 2023).

We elicited inflation perceptions by asking respondents for a point estimate of the inflation rate in their country over the 12 months before the survey. For inflation expectations, we asked respondents for the lowest possible, highest possible, and most likely rate of inflation in their country over the subsequent 12 months. This elicitation format allows us to compute the range as a proxy of uncertainty as in Bachmann et al. (2026), while avoiding cognitively demanding probability-distribution questions (Guiso et al., 2002; Coibion et al., 2024; D’Acunto et al., 2025).¹⁰ We also elicited point expectations for the average nominal wage or salary of all

¹⁰We avoided eliciting probability distributions with pre-specified bins for two additional reasons. First, based on Morning Consult’s trials, such a question would have required us to eliminate five other questions to keep the average response time within the limit allowed by our budget. Second, the bins would have had to be chosen at the country level, which would have complicated cross-country comparisons because the response scale affects elicited moments of inflation expectations (Becker et al., 2023; Boctor et al., 2024).

workers in the respondent’s country, average house-price growth, and interest rates over the subsequent 12 months. In addition, the survey elicited respondents’ saving plans. Unlike macroeconomic expectations, however, these household-level forecasts cannot be compared to ex-post realizations, which we do not observe.

The final part of the survey elicited information acquisition, cognition, preferences, trust, and political alignment. We first asked respondents how important a set of non-mutually-exclusive information sources was for forming their views about inflation. Respondents rated each source on a four-point qualitative scale ranging from “Not important at all” to “Very important.” The sources included, among others, “TV,” “Newspapers,” and “Personal Shopping,” following [D’Acunto et al. \(2021\)](#). We also asked whether respondents thought about the prices of specific goods, such as “eggs,” “gasoline,” or “rent,” when answering the inflation questions. This elicitation follows [D’Acunto et al. \(2021\)](#), who show that consumers often recall concrete goods when forming inflation beliefs. We then used standard questions from the literature to elicit financial and numerical literacy ([Lusardi and Mitchell, 2014](#)), risk and time preferences ([Falk et al., 2018](#)), and trust in a set of country-level and international institutions, including the central bank ([Guiso et al., 2006](#)). Finally, we asked respondents whether the political party that best represented their views was in government, or part of the governing coalition, at the time of the survey ([Binder et al., 2024](#); [Kuang et al., 2024, 2025](#)).¹¹

The survey included two attention checks. The first was a randomly generated arithmetic task asking respondents to solve a simple addition or subtraction problem. The second asked respondents about their use of prominent social-media and media sources and included two non-existent sources, which allowed us to identify respondents who overreported media use.

¹¹Because of the potential sensitivity of these questions, we could not ask the trust in the government and political-alignment questions in Egypt, Saudi Arabia, the United Arab Emirates, Morocco, the People’s Republic of China, the Russian Federation, and Zimbabwe.

3.2 Summary Statistics

Table 1 reports summary statistics of expectations and perceptions, demographics, use of information sources, and trust in institutions at the individual respondent level. We winsorize all continuous perceptions and expectations at the 5% and 95% levels. Table A.1 in the Online Appendix further reports demographic summary statistics by country.

On average, across the 46,285 survey participants, individuals perceive a prevailing inflation rate of 14.70% and expect slightly higher inflation (15.75%) over the subsequent 12 months. Both inflation perceptions and expectations display a fat right tail. We compute two variables to capture the relation between perceptions/expectations and actual realizations: *Inflation Perception Error* is the ratio between a household’s perceived inflation in their country over the previous 12 months and the actual realization, which was public knowledge at the time of our survey;¹² *Inflation Expectation Error* is the ratio between a household’s expected inflation over the 12 months following our survey and the actual realization of the inflation rate in their country 12 months after the survey. We define both variables as ratios rather than first differences to make them more comparable across countries with very different levels of inflation, given that differences are systematically larger in countries with higher inflation levels, and vice versa. We will run all our empirical tests both with and without country fixed effects to assess separately across- and within-country variation. The summary statistics reveal that the vast majority of consumers have positive perception and expectation errors, with ratios larger than 1 at the 34th and 15th percentiles, respectively. On average, survey participants expect house prices to increase by 4% and an average expected change in interest rates of 2.5%.

In most countries, a large percentage of respondents are between 35 and 64 years old. Bangladesh, Egypt, and India are an exception because 70% of the respondents are younger. The percentage of females in the survey is 48% and is equally balanced across countries with the exception of UAE, India, and Nigeria where they represent less than a third of the sam-

¹²We obtain data on realized inflation from official statistics from the OECD or from national statistical offices of the respective countries.

ple. Cross-sectional heterogeneity is present also in terms of education and financial literacy: in Argentina and Hungary more than 70% of survey participants understand basic financial questions, whereas in other countries, such as Canada, India, Japan, Malaysia, Saudi Arabia, and Thailand, the majority of respondents have limited financial knowledge. Also, in most countries, more than half of those surveyed are full-time employed, except for Bangladesh, Hungary, and the United States.

On top of respondent-level variation, our data provide insight on cross-country variation. We display this spatial variation graphically in [Figure 1](#). In both Panels, shaded countries are those that belong to our survey, and the darker is a country, the higher is the variable described in each Panel. In Panel A, we focus on respondents' *Inflation Perception Error*, while in Panel B on the *Inflation Expectation Error*.

First, both Panels document that the average perception and expectation errors are positive in all countries in our sample, which is consistent with evidence in the literature of a zero-lower bound on households' inflation expectations ([Gorodnichenko and Sergeyev, 2021](#)). Second, the ratio is larger than 1 for the vast majority of countries, indicating that even at the country level perceptions and expectations are systematically higher than the actual realizations. Third, both Panels document substantial variation in both perception and expectation errors across countries, with the lowest values in Norway and the highest in China.¹³ Importantly, the ratios are not merely capturing inflation levels: we measure similar values for countries with substantially different inflation levels at the time of our survey. Fourth, we note that variation in perception and expectation errors is not clustered at the regional level, but spatial variation exists also across countries within the same regions, including countries in the same monetary union (Euro Area), where households face the same monetary policy.

Overall, our data display large variation in subjective perceptions, expectations, and errors across both individual households and countries. In the rest of the paper, we investigate the

¹³For robustness, we reran all our results excluding one country at a time, including China, which appears saliently as being the country with the highest perception and expectation errors. All our results, available upon request, are virtually identical when excluding China or any other countries from the sample.

potential role of information sources in explaining this global heterogeneity.

4 Use of Information Sources and Their Quality

We presented in randomized order the eight information sources that households most commonly reported using in U.S.-based surveys (D’Acunto et al., 2021). Based on the taxonomy in Section 2, we classify three sources as aggregate—official government and central-bank reports, TV, and newspapers—and five as local—utility bills, own shopping activities, family members, friends or co-workers, work-related activities such as interactions with customers, suppliers, or competitors, and social media.¹⁴ Respondents were not made aware of this taxonomy. They reported how strongly they relied on each source when forming perceptions and beliefs about the economy on an ordered scale from 1 “very important” to 4 “not important at all.” Respondents were thus free to indicate they relied on all and any sources, and with varied intensity.

Figure 2 reports the share of respondents who declared that each source was either “Very Important” or “Somewhat Important” to them to form economic perceptions and beliefs. Figure 2 reveals substantial heterogeneity in source usage. Two local sources—utility bills and shopping activities—are the most frequently used (more than 80% of respondents). And yet, aggregate sources are also widely used. In particular, about 70% of respondents claim using official reports. The figure highlights two facts that motivate the rest of our analysis. First, local information sources are central to households’ macroeconomic belief formation. Second, reliance on local sources is not simply the mechanical consequence of aggregate information being unavailable, because many respondents also report using aggregate sources.

A potential concern is that some respondents may overreport their use of aggregate information sources, especially official reports. Such overreporting could arise from social desirability,

¹⁴We classify social media as a local source because respondents often encounter discussions of concrete and salient prices, such as grocery and utility prices, on social media. In principle, social media can also transmit aggregate information through official institutional accounts and traditional media outlets’ accounts (Ehrmann and Wabitsch, 2022). Consistent with this ambiguity, the empirical patterns below show that social media resembles other local sources along some dimensions but aggregate sources along others.

if respondents believe that surveyors expect them to rely on official statistics. It could also arise in countries in which respondents fear that the government may observe their answers and penalize them for reporting that they do not use official sources. This concern works against our main empirical predictions. If respondents report using aggregate sources without actually relying on them, their perceptions and expectations should not be systematically more accurate than those of respondents who report using local sources.

4.1 Assessing the Quality of Information Sources: Economic Perceptions

We next assess whether the aggregate vs. local taxonomy captures meaningful differences in the quality of the information households use. Quality is the extent to which information sources provide representative signals about the macroeconomic variables about which agents form beliefs. The key prediction we bring to the data is that respondents who declare relying on aggregate sources form perceptions that are closer to realized macroeconomic outcomes, whereas those who rely on local sources form biased perceptions, as those sources provide systematically unrepresentative signals.

We assess this conjecture by relating the use of each information source to respondents' perceptions and perception errors. [Figure 3](#) reports the results for inflation. Each bar reports the estimated coefficient $\hat{\beta}$ from a separate regression of the following form:

$$Perceptions_i = \alpha + \beta Information\ Source_k + X_i' \delta + \eta_c + \epsilon_i, \quad (1)$$

where $Perceptions_i$ is respondent's i numerical inflation perception in her country over the 12 months before the survey in the left panel, and respondent's i *Inflation Perception Error* in the right panel; $Information\ Source_k$ is a dummy variable that equals 1 if the respondent declares that source k is "Very Important" or "Somewhat Important" for the formation of her subjective expectations for each information source k separately, and zero otherwise.

Demographic controls X_i include dummy variables for respondents' gender, financial literacy, risk tolerance, numeracy, impulsiveness, patience, expertise with financial decisions, grocery shopping frequency, gas purchase frequency, and ability to cover sudden liquidity shocks; η_c are a full set of country fixed effects, which absorbs systematic cross-country differences, including different levels of inflation at the time of the survey. We cluster standard errors at the country level.

Figure 3 supports the conjecture that local information sources are associated with more biased macroeconomic perceptions. The summary statistics revealed that the vast majority of respondents form positively biased inflation perceptions, but those who use local sources (red bars) form more positively biased perceptions than those who use official sources. For instance, a respondent who states that shopping is an important source for her has more than 2 percentage points higher perceived inflation relative to someone who does not consider shopping a relevant information source. In contrast, the perceptions are economically and statistically lower among respondents who use aggregate information sources—about 2 percentage point lower for those who use official reports as a source of information. To gauge economic magnitudes, note that each of these estimated coefficients amounts to about 15% of the average inflation perceptions in the full sample (14.7%). We observe similar patterns for *Inflation Perception Error*. In terms of economic magnitudes, the estimated effects range from about 10% to 16% of the average perception error in the sample.

In Figure 3 and equation (1), we estimate the conditional correlation between using each source and macroeconomic perceptions separately. Because respondents can and do access multiple sources of information, we also estimate specifications that orthogonalize the indicators for access to each information source by adding them all to the right-hand side of the same linear regression. We report the results for inflation perception errors in columns (1)–(3) of Table 2. These multivariate estimates confirm the patterns in Figure 3. Respondents who rely more on aggregate information sources form less biased perceptions, whereas those who rely more on local sources form more biased perceptions. The estimates are similar in the raw data, with controls, when we allow for cross-country variation, as well as when we restrict the

comparison to respondents within the same country.

Overall, these results support the taxonomy based on which aggregate and local sources differ in the representativeness of the signals they provide for aggregate macroeconomic variables. Aggregate sources are associated with perceptions that are closer to realized macroeconomic outcomes—realizations that are public knowledge at the time perceptions are elicited. This result is consistent with the notion that aggregate sources provide more representative information about the relevant aggregate distribution. In contrast, local sources are associated with systematically higher perception biases, which is consistent with the notion that when households draw on salient local signals from their own local economic environments they do not obtain representative signals about the relevant aggregate distribution.

5 Information Sources and Subjective Expectations

5.1 Use of Information Sources and Beliefs: Global Cross Section

We next study how the information sources households use relate to their subjective macroeconomic expectations.

Figure 4 reports the results for the three macroeconomic variables for which we elicit expectations: inflation, house prices, and interest rates. For each variable, we regress numerical expectations on indicators that equal one if the respondent reports using each information source, and zero otherwise. The specifications include demographic characteristics and country fixed effects, which absorb time-invariant country-level averages, such as potential systematic effects due to national economic narratives, institutional environments, and differences in the independence of economic institutions and the press, in addition to different realizations of macroeconomic variables. Hence, we can interpret within-country estimates as differences in expectations relative to the realized aggregate outcome faced by respondents in the same country.

The results for inflation expectations reveal a sharp difference between respondents who

use aggregate information sources and those who use local sources. The inflation expectations of those using official reports are, all else equal, more than 2 percentage points lower than those of otherwise similar respondents who do not consider official reports as important to form their expectations. The associations for TV and newspapers are about half the size and remain statistically significant. In contrast, respondents who use local sources form higher inflation expectations. The relationship is especially pronounced for utility bills and shopping activities, whose users' inflation expectations are almost 2 percentage points higher than those of respondents who do not consider these sources as important. In terms of economic magnitudes, note that the average point estimate for inflation in the full cross-section is 15.3%, and hence the estimates range from about -13% to +13% around the sample average.

The same qualitative patterns appear for the other macroeconomic expectations we elicit. House-price expectations are up to 0.4 percentage points lower among respondents who use aggregate sources and up to 1 percentage point higher among those who use local sources. Despite their smaller sizes, these estimated effects are not less relevant than those for inflation expectations in terms of economic magnitudes: the sample average of house-price expectations is only about 4%, and hence the estimates range from -10% to +25% of the sample average. For expected changes in interest rates, estimates remain statistically significant and economically meaningful for local sources of information. Once again, because the sample average for expected changes in interest rates is 2.48%, most estimated magnitudes are, in fact, relatively larger than for inflation expectations, ranging from about 16% to 28% of the sample mean for local information sources.

Survey respondents might report the mode of inflation expectations when answering the most likely inflation rate question. For robustness purposes, the left panel of [Figure A.1](#) in the Online Appendix shows that the pattern is qualitatively and quantitatively very similar if we consider the average of the maximum and minimum possible values respondents think inflation could reach in their country over the following 12 months.¹⁵ This result dismisses concerns about question framing or reporting inaccurate and rounded point estimates.

¹⁵Due to time constraints, we could only elicit inflation expectations using this question format.

Incidentally, [Figure A.1](#) also reports relevant patterns for proxies of expectations’ uncertainty. The patterns by information sources we document for average inflation expectations arise also for the range of expectations (maximum minus minimum possible values) as well as for households’ tendency to report point estimates of expectations rounded to multiples of five, both of which the literature has interpreted as proxies of expectations’ uncertainty ([Bachmann et al., 2026](#); [Manski and Molinari, 2010](#); [Binder, 2017](#); [Giustinelli et al., 2022](#)). Systematic differences in the information sources households access are thus not only shaping the level of their inflation expectations, but also their uncertainty, which has been shown to affect economic choices independently from the level ([Bloom, 2009, 2014](#); [Coibion et al., 2024](#)).

Similar to the assessment of perceptions, in columns (4)–(6) of [Table 2](#), we report multivariate specifications in which indicators for all information sources enter the same regression. The outcome variable is the *Inflation Expectation Error*. These specifications confirm the patterns in [Figure 4](#). Conditional on demographics and, in the most restrictive specifications, country fixed effects, the use of aggregate sources is associated with lower expectation errors, whereas the use of local sources with higher expectation errors. The estimates are similar in sign to the graphical evidence and support the claim that source usage is strongly related to systematic differences in macroeconomic belief formation.

5.2 Use of Information Sources and Beliefs by Country

The evidence we discussed so far is based on the full cross section of respondents absorbing systematic differences across countries. We next estimate the relationship between information sources and expectations separately by country. This exercise serves two purposes. First, it assesses whether the average relationships we document in [Figure 4](#) are common to many institutional settings or instead driven by outlier countries. Second, it quantifies the cross-country range of the association between information-source use and subjective expectations.

Following [Falk et al. \(2018\)](#), we report the estimated country-level coefficients after sorting countries by the size of the estimate, from smallest to largest. To make the comparison

meaningful, we standardize inflation expectations at the country level. [Figure 5](#) reports the results for inflation expectations. For brevity, we report the corresponding results for interest-rate expectations and house-price expectations in [Figure A.2](#) and [Figure A.3](#) of the Online Appendix. To facilitate interpretation, the figures use marker shape and color to denote statistical significance: red circles, blue diamonds, and pink triangles indicate estimates that are statistically significant at the 1%, 5%, and 10% levels, respectively, whereas green hollow circles denote estimates for which we cannot reject the null at standard levels of significance.

[Figure 5](#) shows that the baseline estimates' signs hold separately for most countries in the sample, and hence are not driven by specific country clusters or outliers. In Panel A, the country-level relationship between accessing official reports and inflation expectations is negative in 38 out of 47 countries, and statistically significant in many countries. Similar qualitative patterns arise for the other aggregate sources.

Panel B reports the country-level estimates for local sources. The pattern is approximately symmetric: the estimated associations are positive in most countries. As in the full-sample analysis, the relationship is strongest for energy bills and shopping activities, that is, the local sources that most directly expose households to salient and volatile price changes. The estimates for other local sources are flatter across countries, but their signs remain broadly consistent with the aggregate evidence.

We detect similar patterns by countries for house-price expectations and interest-rate expectations in [Figure A.2](#) and [Figure A.3](#) of the Online Appendix. In particular, the use of local information sources is almost universally related to higher expectations for both sets of prices.

Overall, the country-level OLS estimates show that the relationship between information-source use and subjective expectations is not specific to a small set of economies or countries facing peculiar conditions. Reliance on aggregate or local sources of economic information predicts the direction of belief differences and biases across a wide range of countries.

5.3 Intensity of Exposure to Information Sources and Beliefs

The associations discussed so far concern the extensive margin: whether respondents use a given information source. Unobserved consumer-level characteristics that drive source selection might in principle also directly shape beliefs. To assuage this concern, we also study the intensive margin of exposure to information sources, whereby we only compare consumers who access the same information source, but with different intensities. For this purpose, we collected quantitative measures of how frequently respondents engage with two local sources: grocery shopping experiences and gas station visits.¹⁶

Figure 6 reports the results. Panel A restricts the sample to respondents who report using their own shopping activities as a source of economic information. Panel B restricts the sample to respondents who report using utility bills as a source of information.¹⁷ In each panel, the y-axis plots respondents' *Inflation Expectation Error*. The x-axis sorts respondents by the reported frequency of exposure to each local source.

For both sources, the intensity of exposure is positively associated with the size of the inflation expectation error. This pattern is consistent with the conjecture that when using local sources of information, greater exposure and hence a higher information acquisition can amplify rather than attenuate expectation biases, which is what standard rational inattention frameworks would predict.

6 Demographics and Beliefs: Differences in Belief Formation vs. Information Sources

An ideal experiment to estimate the causal effect of information sources on macroeconomic expectations would randomly assign consumers to different information environments for a sustained period of time, restrict access to alternative sources, and then elicit their beliefs. Such

¹⁶Unfortunately, we could not ask questions to elicit the intensity of attendance to all information sources. We thus focused on the two sources that predict the largest positive biases in inflation perceptions and expectations.

¹⁷Figure A.4 in the appendix reports equivalent figures for the ratio of perceived to realized inflation.

an experiment is not feasible in our setting and, to our knowledge, has not been implemented in comparable settings. Our analysis therefore does not aim to identify the causal effect of information-source exposure. Instead, we use the global cross section to assess whether information-source choice helps account for demographic differences in macroeconomic beliefs documented in earlier work.

Specifically, we ask whether macroeconomic expectations differ across demographic groups conditional on the information sources respondents use. If demographic groups form similar expectations after conditioning on information-source choice, then demographic characteristics may matter primarily because they predict sorting into different information environments.¹⁸ In contrast, if large demographic differences remain within the same information-source categories, this would suggest that demographic groups process similar information differently. This distinction is important because much of the existing evidence on demographic heterogeneity in expectations is consistent with either interpretation, but the two possibilities have very different implications for models of belief formation. They also have different policy implications, for instance, regarding the role of communication as a policy tool.

Figure 7 compares graphically the relationship between information sources and respondents' *Inflation Expectation Error*. Each panel replicates the baseline relationship between information-source use and inflation expectations from Figure 4, but estimates it separately across demographic splits of respondents. The exercise therefore compares whether the association between source use and inflation expectations differs across groups defined by observable demographic characteristics.

The results show that the relationship between information sources and inflation expectations is remarkably similar across demographic groups. Both qualitatively and quantitatively, respondents who use aggregate sources report lower inflation expectations, whereas respondents who use local sources report higher inflation expectations, irrespective of the demographic split considered. Conditional on information-source use, demographic differences in expectation for-

¹⁸We provide graphical evidence on the sorting in local and aggregate information sources by demographics in Figure A.5 and Figure A.6.

mation appear substantially smaller than the unconditional differences documented in earlier work.

But then, given the mounting single-country evidence of systematic differences in expectation biases by demographic groups, do we observe systematic differences in the use of information sources? Indeed, we do. To reduce the dimensionality of our reported results, we compute the share of local sources and the share of aggregate sources that each consumer reports as being relevant to the formation of their beliefs. We compare these two outcomes in each country in the sample and across demographics that earlier research associates with higher or lower expectation biases. For brevity, we report the evidence in [Figure A.5](#) and [Figure A.6](#) of the Online Appendix.

We find that women, main grocery shoppers for their households, frequent gas purchasers, married, impulsive, and impatient consumers all are systematically more likely to sort into local information sources than other demographics in virtually all countries in our sample. In contrast, college educated individuals, including those with economics-related degrees, the financially literate, those with high numeracy, and the employed are more likely to select into aggregate sources of economic information relative to other demographics.

Overall, our global facts suggest that the systematic variation in expectation errors by demographic groups are largely due to the fact that different demographics sort systematically into different information sources. In contrast, conditional on accessing the same information sources, consumers with different demographics form strikingly similar expectation errors.

These results also inform advances in the modeling of heterogeneous expectations. They suggest that heterogeneity in the information environment may be a first-order source of expectation biases, and potentially more important than demographic heterogeneity in processing the same signals and economic information.

7 Distrust in Economic Institutions

Information sources appear to be important in explaining cross-sectional differences in macroeconomic perceptions, expectations, and biases, even above and beyond consumers' demographic characteristics. What drives households' sorting across alternative information sources is a crucial yet open question. As we discussed above, systematic differences in the search costs and cognitive costs of accessing aggregate vs. local sources seem an implausible explanation because, if anything, both sets of costs should be lower for aggregate sources for all demographics.

We propose and assess empirically an alternative potential mechanism—consumers' extent of distrust in the economic institutions that produce and disseminate aggregate economic information.

Governments (and their statistical agencies) and central banks produce and disseminate the aggregate statistics households can access to obtain macroeconomic information. At the same time, they influence macroeconomic outcomes with their policy measures. If households trust these institutions, they may view aggregate information sources as credible and informative. If they distrust them, they might reject the information they produce. For instance, households might be concerned that aggregate information is plagued by institutions' conflicts of interest, whereby governments and central banks have the incentive to manipulate aggregate information to preserve their political support and/or to support the implementation of unpopular policy interventions. Distrustful households would thus rely more on the only information sources they can access and witness directly, without the intermediation of economic institutions—local sources of information.

To test this mechanism, we elicited respondents' distrust in their country's central bank and government separately, as well as trust in other institutions.¹⁹ We compute the first principal component of distrust in governments and central banks and use it as a summary

¹⁹We were not able to ask questions about trust in national governments in Egypt, Saudi Arabia, the United Arab Emirates, Morocco, the People's Republic of China, the Russian Federation, and Zimbabwe. For this reason, these countries are excluded from the analysis in this section. Results are virtually unchanged if we focus on trust in the central bank and focus on the full set of countries.

measure of distrust in public economic institutions. This measure captures the extent to which respondents view the main producers of aggregate economic information as untrustworthy.

Figure 8 reports country-level estimates of the relationship between distrust in public economic institutions and information-source use. For each country and each source, we regress distrust in the central bank and government on an indicator for whether the respondent uses a given information source. We then sort the country-level coefficients from smallest to largest, following the presentation used in the earlier country-level analyses. Panel A shows that distrust is strongly negatively associated with the use of aggregate information sources in all countries, and statistically significant in all but three instances. Note that the question about trust in institutions and that on information sources were asked in very different parts of the survey and after respondents thought about topics unrelated to economic perceptions, beliefs, and information sources. It is implausible that households purposefully manipulated their trust in the government and in the central bank to be consistent with their earlier reported usage of official sources of economic information.

Panel B reports the corresponding estimates for local information sources. The estimates are both negative and positive, and statistical significance is more sparse, especially for shopping activities and energy bills. These patterns suggest that distrust in the government and central bank are strongly negatively associated with the use of aggregate information sources. In contrast, households access local economy sources irrespective of their distrust in economic institutions.

We then ask whether trust also helps explain perception and expectation biases. Figure 9 provides univariate evidence. The top panels plot binscatter relationships between distrust in public economic institutions and inflation perception errors (left) and inflation expectation errors (right). Both relationships are strongly positive: respondents who distrust their government and central bank more report inflation perceptions and expectations that are farther above realized inflation. The bottom panels show the corresponding distributional patterns. Higher distrust shifts the distribution of perception and expectation biases to the right, indicating that the relationship is not driven only by changes in the mean.

Because our data are a single global cross section, we do not have quasi-exogenous variation in trust in public economic institutions. We therefore complement the baseline evidence with placebo-style comparisons that help assess whether the relationship reflects trust in the producers of aggregate economic information rather than generalized distrust or political alignment. Specifically, we compare trust in governments and central banks with trust in private economic institutions—businesses and corporations, and the stock market. Trust in public and private economic institutions may both capture broad attitudes toward the economic system. However, only governments and central banks produce, disseminate, and are politically accountable for the aggregate macroeconomic information that households can use to evaluate inflation, interest rates, and other aggregate outcomes. We also use respondents’ answers on whether the political party closest to their views was in government, or part of the governing coalition, at the time of the survey. This measure helps distinguish distrust in public economic institutions from general political opposition to the incumbent government.

Table 3 reports multivariate specifications in which we regress individual-level perception and expectation biases on distrust in public economic institutions, distrust in private economic institutions, political alignment, and the demographic controls used throughout the paper. We estimate specifications both with and without country fixed effects. The table shows two main patterns. First, distrust in public economic institutions remains economically and statistically significantly associated with both perception and expectation biases after controlling for trust in private economic institutions, political alignment, and demographics. The magnitude of the association is similar in specifications that exploit cross-country variation and in specifications that compare respondents within the same country, indicating that within-country heterogeneity in trust is central to the relationship. Second, distrust in private economic institutions and political alignment have little explanatory power for perception biases and especially for expectation biases once distrust in governments and central banks is included. These patterns support the interpretation that the relevant dimension is not generalized distrust or political disagreement per se, but distrust in the institutions that produce and disseminate aggregate economic information.

7.1 Distrust in Economic Institutions and Past Economic Conditions

Once again, distrust in governments and central banks is endogenous. The results in the previous section show that observed demographics and time-invariant country characteristics do not fully account for the relationship between distrust, information-source choice, and belief biases. The estimated associations remain economically and statistically significant when we compare respondents within the same country.

At the same time, our cross-sectional survey does not allow us to identify the full set of individual-level determinants of distrust. Understanding these determinants is an important direction for future work, especially for models that incorporate trust in economic institutions as a determinant of the perceived quality of aggregate information.

Our setting allows us to assess a potential driver of distrust in economic institutions—a country’s historical macroeconomic conditions.²⁰ Because past macroeconomic realizations are common to all respondents in a country, they cannot explain the within-country variation in distrust documented above. They can, however, help account for cross-country differences in average distrust.

We focus on two dimensions of past inflation dynamics: the average inflation rate over the 10 years before the survey and the standard deviation of inflation over the same period.²¹ The average captures the level of past inflation, whereas the standard deviation captures macroeconomic instability. This distinction allows us to ask whether distrust is associated with a history of high inflation per se or, instead, with a history of volatile economic conditions.

Table 4 reports the results. We regress individual-level distrust in public economic institutions on the country’s average inflation rate over the previous 10 years in column (1), the

²⁰Closely related evidence suggests that individuals update their trust in government based on experienced macroeconomic performance. Using harmonized survey data covering 166 countries, Besley et al. (2026) show that individuals who experienced stronger economic growth over their lifetimes report greater trust in government.

²¹We use 10 years of year on year CPI inflation at the monthly frequency to calculate both the mean and volatility of inflation. Variations in the frequency or the lookback period such as 20 years are largely immaterial for our results.

standard deviation of inflation over the previous 10 years in column (2), both variables jointly in column (3), and the coefficient of variation, defined as the ratio of the standard deviation to the average inflation rate, in column (4). All specifications include demographic controls.²²

Table 4 reveals two main patterns. First, the average level of past inflation is not systematically related to distrust in public economic institutions. The estimated association is economically small and statistically insignificant, both when average inflation enters the regression alone and when it enters jointly with inflation volatility. Second, inflation volatility is positively associated with distrust. The standard deviation of inflation over the previous 10 years is economically and statistically significant, and the estimate remains similar when we control for average inflation. The coefficient of variation delivers the same conclusion: countries with more volatile historical economic conditions display higher average distrust in economic institutions.

Overall, the evidence suggests that distrust is closely related to macroeconomic volatility. A history of high but stable inflation does not appear to predict distrust in economic institutions in our data. Instead, distrust is higher in countries in which economic conditions have been more unstable. This pattern is consistent with the idea that volatile macroeconomic environments undermine the credibility of aggregate economic information and may thereby affect the sources households use to form macroeconomic beliefs.

8 Conclusions

Using novel data on about 47,000 consumers across 47 countries representing 90% of global GDP, we document a set of global facts about how households form macroeconomic perceptions and expectations. The facts help explain why consumers' macroeconomic beliefs are often systematically biased, a regularity documented across countries and over time but difficult to reconcile with models in which agents either observe unbiased signals or become more accurate

²²Past macroeconomic realizations vary only at the country level. We therefore cannot include country fixed effects in these specifications.

when they acquire more information.

Our central finding is that information acquisition is not sufficient for accurate macroeconomic beliefs. The source of information matters. Many consumers rely on signals from their local economic environments—utility bills, shopping chores, family and friends, work-related interactions, and social media—rather than on aggregate sources such as official reports by governments and central banks or traditional media. These local signals are salient but not necessarily representative of the aggregate macroeconomic variables consumers are asked to assess. Consumers who rely on local information sources form more biased macroeconomic perceptions and expectations, and greater exposure to such sources is associated with larger biases.

We also show that trust in public economic institutions is closely related to information-source choice. Governments and central banks produce and disseminate much of the aggregate economic information households could use to evaluate macroeconomic conditions. Consumers who distrust these institutions rely less on aggregate sources. Our setting does not allow us to identify all individual-level determinants of trust, which remains an important topic for future research and for the microfoundation of models of macroeconomic belief formation. At the country level, however, distrust is more strongly associated with the volatility of past inflation than with the average level of past inflation, suggesting that unstable macroeconomic environments may erode the credibility of aggregate economic information.

The facts we document point to a tractable extension of rational-inattention and dispersed-information frameworks. Models should distinguish not only between more and less information, or between more and less precise signals, but also between signals that differ in representativeness and perceived credibility. In such a framework, consumers choose among multiple sources of information whose objective informativeness and subjective trustworthiness may diverge. Trust in the producer of aggregate information can then affect perceived signal quality and, therefore, the willingness of households to rely on aggregate rather than local information. This extension can help account not only for the dispersion of macroeconomic expectations, but also for their systematic bias.

The results also have implications for communication policy. Making aggregate information publicly available may not be enough if households distrust the institutions that produce it or if local signals dominate their lived experience. Effective communication by governments and central banks therefore requires not only clear messages, but also sustained efforts to build credibility with the broader public, especially with groups that are more likely to rely on local information sources ([D’Acunto et al., 2021](#)). Greater trust may induce households to use more representative aggregate information when forming macroeconomic beliefs and thereby strengthen the transmission of fiscal and monetary policy ([D’Acunto et al., 2021](#)).

References

- Afrouzi, H. and C. Yang (2021). Dynamic rational inattention and the phillips curve. CESifo Working Paper 8840, CESifo.
- Andre, P., C. Pizzinelli, C. Roth, and J. Wohlfart (2022). Subjective models of the macroeconomy: Evidence from experts and representative samples. *The Review of Economic Studies* 89(6), 2958–2991.
- Anesti, N., V. Esady, and M. Naylor (2024). *Food prices matter most: Sensitive household inflation expectations*. London School of Economics and Political Science.
- Angeletos, G.-M. and Z. Huo (2021). Myopia and anchoring. *American Economic Review* 111(4), 1166–1200.
- Angeletos, G.-M., Z. Huo, and K. A. Sastry (2021). Imperfect macroeconomic expectations: Evidence and theory. *NBER Macroeconomics Annual* 35(1), 1–86.
- Angeletos, G.-M. and J. La’O (2009). Incomplete information, higher-order beliefs and price inertia. *Journal of Monetary Economics* 56(S), S19–S37.
- Angeletos, G.-M. and C. Lian (2018). Forward guidance without common knowledge. *American Economic Review* 108(9), 2477–2512.
- Angeletos, G.-M. and C. Lian (2023). Dampening general equilibrium: incomplete information and bounded rationality. In *Handbook of economic expectations*, pp. 613–645. Elsevier.
- Angeletos, G.-M. and A. Pavan (2007). Efficient use of information and social value of information. *Econometrica* 75(4), 1103–1142.
- Armantier, O., W. Bruine de Bruin, S. Potter, G. Topa, W. Van Der Klaauw, and B. Zafar (2013). Measuring inflation expectations. *Annual Review of Economics* 5, 273–301.
- Bachmann, R., K. Carstensen, M. Schneider, and M. Menkhoff (2026). Why does firm uncertainty rise in recessions.
- Bailey, M., R. Cao, T. Kuchler, and J. Stroebel (2018). The economic effects of social networks: Evidence from the housing market. *Journal of Political Economy* 126(6), 2224–2276.

- Becker, C., P. Duersch, and T. A. Eife (2023). Measuring inflation expectations: How the response scale shapes density forecasts. Technical report.
- Besley, T., C. Dann, and S. Dray (2026). Growth experiences and trust in government. *The Quarterly Journal of Economics* 141(2), 1761–1822.
- Bianchi, F., C. Ilut, and H. Saijo (2024). Diagnostic business cycles. *Review of Economic Studies* 91(1), 129–162.
- Binder, C. and C. Makridis (2022). Stuck in the seventies: gas prices and consumer sentiment. *Review of Economics and Statistics* 104(2), 293–305.
- Binder, C. C. (2017). Measuring uncertainty based on rounding: New method and application to inflation expectations. *Journal of Monetary Economics* 90, 1–12.
- Binder, C. C., R. Kamdar, and J. M. Ryngaert (2024). Partisan expectations and covid-era inflation. *Journal of Monetary Economics* 148, 103649.
- Blinder, A. S., M. Ehrmann, J. De Haan, and D.-J. Jansen (2024). Central bank communication with the general public: Promise or false hope? *Journal of Economic Literature* 62(2), 425–457.
- Bloom, N. (2009). The impact of uncertainty shocks. *econometrica* 77(3), 623–685.
- Bloom, N. (2014). Fluctuations in uncertainty. *Journal of economic Perspectives* 28(2), 153–176.
- Boctor, V. R., O. Coibion, Y. Gorodnichenko, and M. Weber (2024). On eliciting subjective probability distributions of expectations. Technical report.
- Bordalo, P., N. Gennaioli, and A. Shleifer (2022). Overreaction and diagnostic expectations in macroeconomics. *Journal of Economic Perspectives* 36(3), 223–244.
- Bruine de Bruin, W., W. Vanderklaauw, J. S. Downs, B. Fischhoff, G. Topa, and O. Armantier (2010). Expectations of inflation: The role of demographic variables, expectation formation, and financial literacy. *Journal of Consumer Affairs* 44(2), 381–402.
- Bryan, M. F. and G. Venkatu (2001). The demographics of inflation opinion surveys. *Economic Commentary* (10/15/2001).

- Carroll, C. D. (2003). Macroeconomic expectations of households and professional forecasters. *Quarterly Journal of economics* 118(1), 269–298.
- Cavallo, A., G. Cruces, and R. Perez-Truglia (2016). Learning from potentially biased statistics. *Brookings Papers on Economic Activity* (1), 59–108.
- Cavallo, A., G. Cruces, and R. Perez-Truglia (2017). Inflation expectations, learning, and supermarket prices: Evidence from survey experiments. *American Economic Journal: Macroeconomics* 9(3), 1–35.
- Chahrour, R. and G. Gaballo (2021). Learning from house prices: Amplification and business fluctuations. *Review of Economic Studies* 88(4), 1720–1759.
- Chahrour, R., K. Nimark, and S. Pitschner (2021). Sectoral media focus and aggregate fluctuations. *American Economic Review* 111(12), 3872–3922.
- Christelis, D., D. Georgarakos, T. Jappelli, and M. van Rooij (2020). Trust in the central bank and inflation expectations. *International Journal of Central Banking* 16(6), 1–37.
- Coibion, O., D. Georgarakos, Y. Gorodnichenko, G. Kenny, and M. Weber (2024). The effect of macroeconomic uncertainty on household spending. *American Economic Review* 114(3), 645–677.
- Coibion, O. and Y. Gorodnichenko (2015, January). Is the phillips curve alive and well after all? inflation expectations and the missing disinflation. *American Economic Journal: Macroeconomics* 7(1), 197–232.
- Coibion, O., Y. Gorodnichenko, and M. Weber (2020). Political polarization and expected economic outcomes. Technical report.
- Coibion, O., Y. Gorodnichenko, and M. Weber (2022). Monetary policy communications and their effects on household inflation expectations. *Journal of Political Economy* 130(6), 1537–1584.
- Conrad, C., Z. Enders, and A. Glas (2022). The role of information and experience for households’ inflation expectations. *European Economic Review* 143, 104015.
- Crump, R. K., S. Eusepi, A. Tambalotti, and G. Topa (2022). Subjective intertemporal substitution. *Journal of Monetary Economics* 126, 118–133.

- D’Acunto, F., J. Gao, L. Liu, K. Lu, Z. Wang, and J. Yang (2025). Subjective expectations and financial intermediation. *Working Paper*.
- D’Acunto, F., U. Malmendier, and M. Weber (2023). What do the data tell us about inflation expectations? In *Handbook of economic expectations*, pp. 133–161. Elsevier.
- D’Acunto, F. and M. Weber (2024). Why survey-based subjective expectations are meaningful and important. *Annual Review of Economics* 16, 329–357.
- D’Acunto, F. and M. Weber (2026). Behavioral macroeconomics: Introducing realistic agents in macroeconomics. In *Oxford Research Encyclopedia of Economics and Finance*. Oxford University Press.
- De Bruin, W. B., C. F. Manski, G. Topa, and W. Van Der Klaauw (2011). Measuring consumer uncertainty about future inflation. *Journal of Applied Econometrics* 26(3), 454–478.
- De Bruin, W. B., W. Van Der Klaauw, and G. Topa (2011). Expectations of inflation: The biasing effect of thoughts about specific prices. *Journal of Economic Psychology* 32(5), 834–845.
- De Fiore, F., A. Maurin, A. Mijakovic, and D. Sandri (2024). Monetary policy in the news: communication pass-through and inflation expectations.
- Dietrich, A., E. S. Knotek, R. Schoenle, M. Weber, and K. O. R. Myrseth (2022). Greater than the sum of the parts: Aggregate vs. aggregated inflation expectations. *FRB of Cleveland Working Paper No. 22-20*.
- Dräger, L. and M. J. Lamla (2024). Consumers’ macroeconomic expectations. *Journal of Economic Surveys* 38(2), 427–451.
- Dräger, L., M. Marenčák, G. Nghiem, and M. Paloviita (2026). Heterogeneity in consumers’ economic expectations across euro area countries.
- D’Acunto, F., E. Charalambakis, D. Georgarakos, G. Kenny, J. Meyer, and M. Weber (2024). Household inflation expectations: An overview of recent insights for monetary policy.
- D’Acunto, F., A. Fuster, and M. Weber (2021). Diverse policy committees can reach under-represented groups. Technical report.

- D’Acunto, F., D. Hoang, M. Paloviita, and M. Weber (2021). Human frictions in the transmission of economic policies. Technical report.
- D’Acunto, F., D. Hoang, M. Paloviita, and M. Weber (2022). Iq, expectations, and choice. *The Review of Economic Studies*, rdac075.
- D’Acunto, F., U. Malmendier, J. Ospina, and M. Weber (2021). Exposure to grocery prices and inflation expectations. *Journal of Political Economy* 129(5), 1615–1639.
- D’Acunto, F., U. Malmendier, and M. Weber (2021). Gender roles produce divergent economic expectations. *Proceedings of the National Academy of Sciences* 118(21), e2008534118.
- D’Acunto, F. and M. Weber (2023). Memory and Beliefs: Evidence from the Field. *Working paper*.
- Ehrmann, M. (2025). Trust in central banks. *Journal of Economic Surveys*.
- Ehrmann, M., D. Georgarakos, and G. Kenny (2025). Credibility gains from central bank communication with the public. *European Economic Review* 177, 105069.
- Ehrmann, M. and A. Wabitsch (2022). Central bank communication with non-experts—a road to nowhere? *Journal of Monetary Economics* 127, 69–85.
- Enke, B. (2020). What you see is all there is. *The Quarterly Journal of Economics* 135(3), 1363–1398.
- Faia, E., A. Fuster, V. Pezone, and B. Zafar (2024). Biases in information selection and processing: Survey evidence from the pandemic. *Review of Economics and Statistics* 106(3), 829–847.
- Falk, A., A. Becker, T. Dohmen, B. Enke, D. Huffman, and U. Sunde (2018). Global evidence on economic preferences. *The quarterly journal of economics* 133(4), 1645–1692.
- Fofana, S., P. Patzelt, and R. Reis (2025). *Household disagreement about expected inflation*. Edward Elgar Publishing.
- Fuster, A., R. Perez-Truglia, M. Wiederholt, and B. Zafar (2022). Expectations with endogenous information acquisition: An experimental investigation. *Review of Economics and Statistics* 104(5), 1059–1078.

- Gaballo, G. (2016). Rational inattention to news: The perils of forward guidance. *American Economic Journal: Macroeconomics* 8(1), 42–97.
- Gennaioli, N. and A. Shleifer (2010). What comes to mind. *The Quarterly Journal of Economics* 125(4), 1399–1433.
- Gennaioli, N. and A. Shleifer (2018). *A Crisis of Beliefs: Investor Psychology and Financial Fragility*. Princeton, NJ: Princeton University Press.
- Giustinelli, P., C. F. Manski, and F. Molinari (2022). Tail and center rounding of probabilistic expectations in the health and retirement study. *Journal of Econometrics* 231(1), 265–281.
- Gorodnichenko, Y. and D. Sergeyev (2021). Zero lower bound on inflation expectations. Technical report, National Bureau of Economic Research.
- Guiso, L., T. Jappelli, and L. Pistaferri (2002). An empirical analysis of earnings and employment risk. *Journal of Business & Economic Statistics* 20(2), 241–253.
- Guiso, L., P. Sapienza, and L. Zingales (2006). Does culture affect economic outcomes? *Journal of Economic perspectives* 20(2), 23–48.
- Guiso, L., P. Sapienza, and L. Zingales (2008). Trusting the stock market. *the Journal of Finance* 63(6), 2557–2600.
- Haaland, I., C. Roth, and J. Wohlfart (2023). Designing information provision experiments. *Journal of economic literature* 61(1), 3–40.
- Hajdini, I., E. S. Knotek II, J. Leer, M. Pedemonte, R. Rich, and R. Schoenle (2024). Indirect consumer inflation expectations: Theory and evidence. *Journal of Monetary Economics* 145, 103568.
- Hébert, B. M. and J. La’O (2023). Information acquisition, efficiency, and nonfundamental volatility. *Journal of Political Economy* 131(10), 2666–2723.
- Jonung, L. (1981). Perceived and expected rates of inflation in sweden. *The American Economic Review* 71(5), 961–968.
- Kamdar, R. and W. Ray (2022). *Polarized expectations, polarized consumption*. SSRN.

- Kindermann, F., J. Le Blanc, M. Piazzesi, and M. Schneider (2021). Learning about housing cost: Survey evidence from the german house price boom. Technical report, National Bureau of Economic Research.
- Kohlhas, A. N. and A. Walther (2021). Asymmetric attention. *American Economic Review* 111(9), 2879–2925.
- Kuang, P., M. Weber, and S. Xie (2024). Perceived political bias of the federal reserve. Technical report.
- Kuang, P., M. Weber, and S. Xie (2025, February). Central bank communication with the polarized public. Working Paper 33524.
- Kuchler, T., M. Piazzesi, and J. Stroebl (2023). Housing market expectations. In *Handbook of economic expectations*, pp. 163–191. Elsevier.
- Kuchler, T. and B. Zafar (2019). Personal experiences and expectations about aggregate outcomes. *The Journal of Finance* 74(5), 2491–2542.
- Lamla, M. J. and S. M. Lein (2014). The role of media for consumers’ inflation expectation formation. *Journal of Economic Behavior & Organization* 106, 62–77.
- Lamla, M. J. and D. V. Vinogradov (2019). Central bank announcements: Big news for little people? *Journal of Monetary Economics* 108, 21–38.
- Larsen, V. H., L. A. Thorsrud, and J. Zhulanova (2021). News-driven inflation expectations and information rigidities. *Journal of Monetary Economics* 117, 507–520.
- L’Huillier, J.-P., S. R. Singh, and D. Yoo (2024). Incorporating diagnostic expectations into the new keynesian framework. *Review of Economic Studies* 91(5), 3013–3046.
- Lucas, R. E. (1975). An equilibrium model of the business cycle. *Journal of Political Economy* 83(6), 1113–1144.
- Lusardi, A. and O. S. Mitchell (2014). The economic importance of financial literacy: Theory and evidence. *Journal of Economic Literature* 52(1), 5–44.
- Maćkowiak, B., F. Matějka, and M. Wiederholt (2023). Rational inattention: A review. *Journal of Economic Literature* 61(1), 226–273.

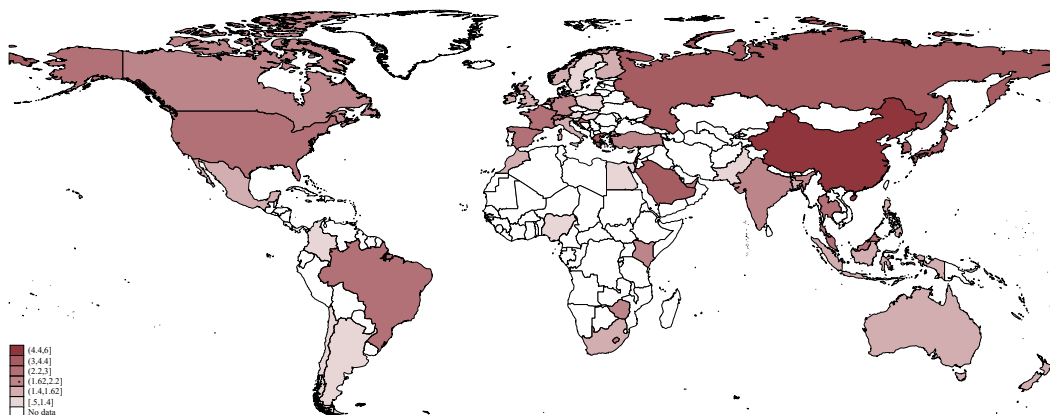
- Maćkowiak, B. and M. Wiederholt (2009). Optimal sticky prices under rational inattention. *American Economic Review* 99(3), 769–803.
- Mankiw, N. G., R. Reis, and J. Wolfers (2004). Disagreement about inflation expectations. In M. Gertler and K. Rogoff (Eds.), *NBER Macroeconomics Annual 2003, Volume 18*, pp. 209–270. Cambridge, MA: MIT Press.
- Manski, C. F. and F. Molinari (2010). Rounding probabilistic expectations in surveys. *Journal of Business & Economic Statistics* 28(2), 219–231.
- Morris, S. and H. S. Shin (2002). Social value of public information. *American Economic Review* 92(5), 1521–1534.
- Nimark, K. (2008). Dynamic pricing and imperfect common knowledge. *Journal of Monetary Economics* 55(2), 365–382.
- Paciello, L. and M. Wiederholt (2014). Exogenous information, endogenous information, and optimal monetary policy. *Review of Economic Studies* 81(1), 356–388.
- Patzelt, P. and R. Reis (2024). Estimating the rise in expected inflation from higher energy prices. Technical report.
- Pfajfar, D. and E. Santoro (2009). Asymmetries in inflation expectations across sociodemographic groups. Technical report.
- Reis, R. (2006). Inattentive consumers. *Journal of monetary Economics* 53(8), 1761–1800.
- Roth, C., S. Settele, and J. Wohlfart (2022). Risk exposure and acquisition of macroeconomic information. *American Economic Review: Insights* 4(1), 34–53.
- Sims, C. A. (2003). Implications of rational inattention. *Journal of Monetary Economics* 50(3), 665–690.
- Weber, M., F. D’Acunto, Y. Gorodnichenko, and O. Coibion (2022). The subjective inflation expectations of households and firms: Measurement, determinants, and implications. *Journal of Economic Perspectives* 36(3), 157–184.
- Weber, M., Y. Gorodnichenko, and O. Coibion (2023). The expected, perceived, and realized inflation of us households before and during the covid19 pandemic. *IMF Economic Review* 71(1), 326–368.

Wehrhöfer, N. (2026). Energy prices and inflation expectations: Evidence from households and firms. *Journal of Political Economy Macroeconomics* 4(1), 135–163.

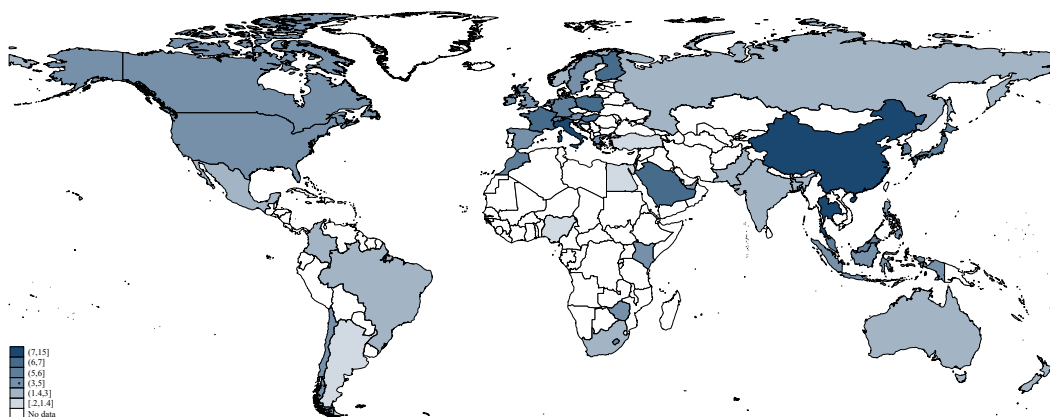
Woodford, M. (2003). Imperfect common knowledge and the effects of monetary policy. *Knowledge, information, and expectations in modern macroeconomics: In honor of Edmund S. Phelps* 25(1), 4.

Figure 1: Global Macroeconomic Perception and Expectation Errors

Panel A. Average Inflation Perception Error

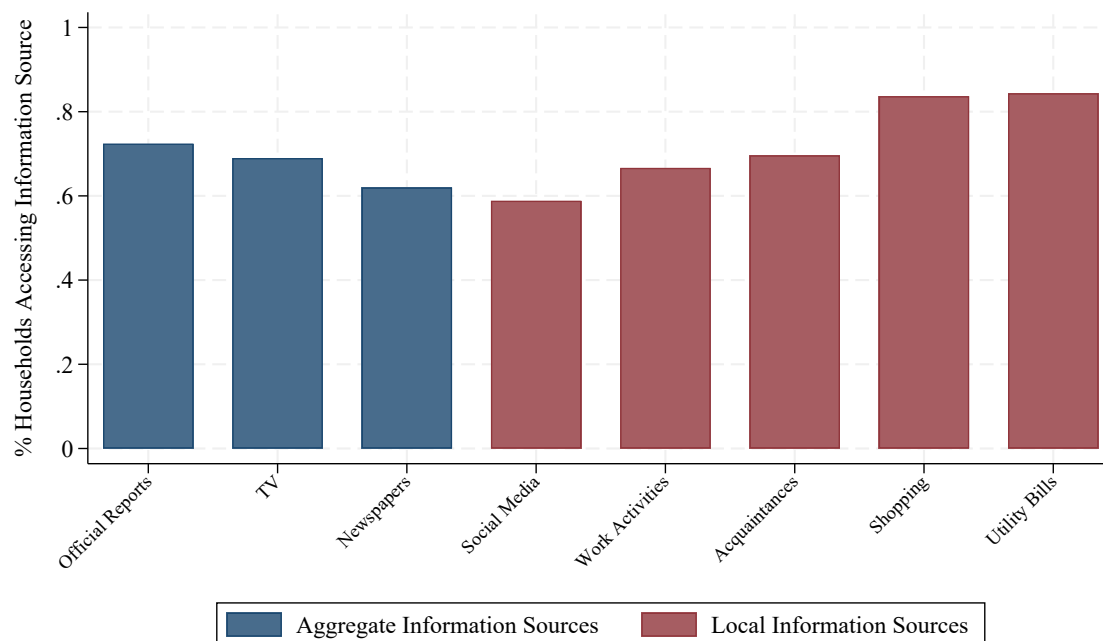


Panel B. Average Inflation Expectation Error



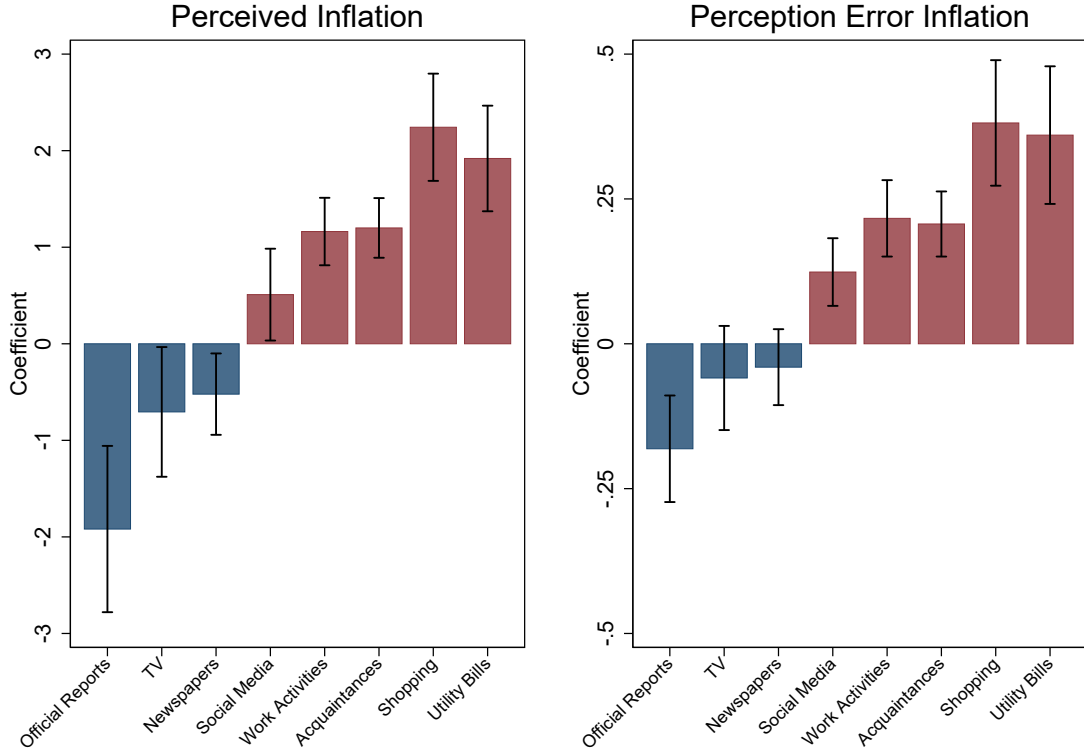
Notes: This figure plots the country-level averages of the inflation perception error (top map) and inflation expectation error (bottom map) based on 46,285 respondents across 47 countries in the *Global Subjective Expectations Survey* conducted in April/May 2023. The respondents were asked to provide the perceived inflation in their country in the previous 12-month period. Inflation perception error is the ratio of perceived year-to-year inflation over actual realized yearly inflation in April 2023. Inflation expectation error is the ratio of expected inflation over the following 12 months over actual realized inflation in April 2024. Both variables are winsorized at 5-95 levels.

Figure 2: **Distribution of Use of Aggregate vs. Local Economic Information Sources**



Notes. This figure plots the reported importance of different information sources for the formation of inflation expectations based on the survey responses of 46,285 respondents representative of the households across 47 countries in April/May 2023 from the *Global Subjective Inflation Expectations Survey*. The bars represent the fraction of respondents who consider an information source as “Very Important,” or “Somewhat Important” based on survey question 25. The classification of information sources follows the taxonomy in [section 2](#).

Figure 3: **Quality of Information Sources: Inflation Perceptions**

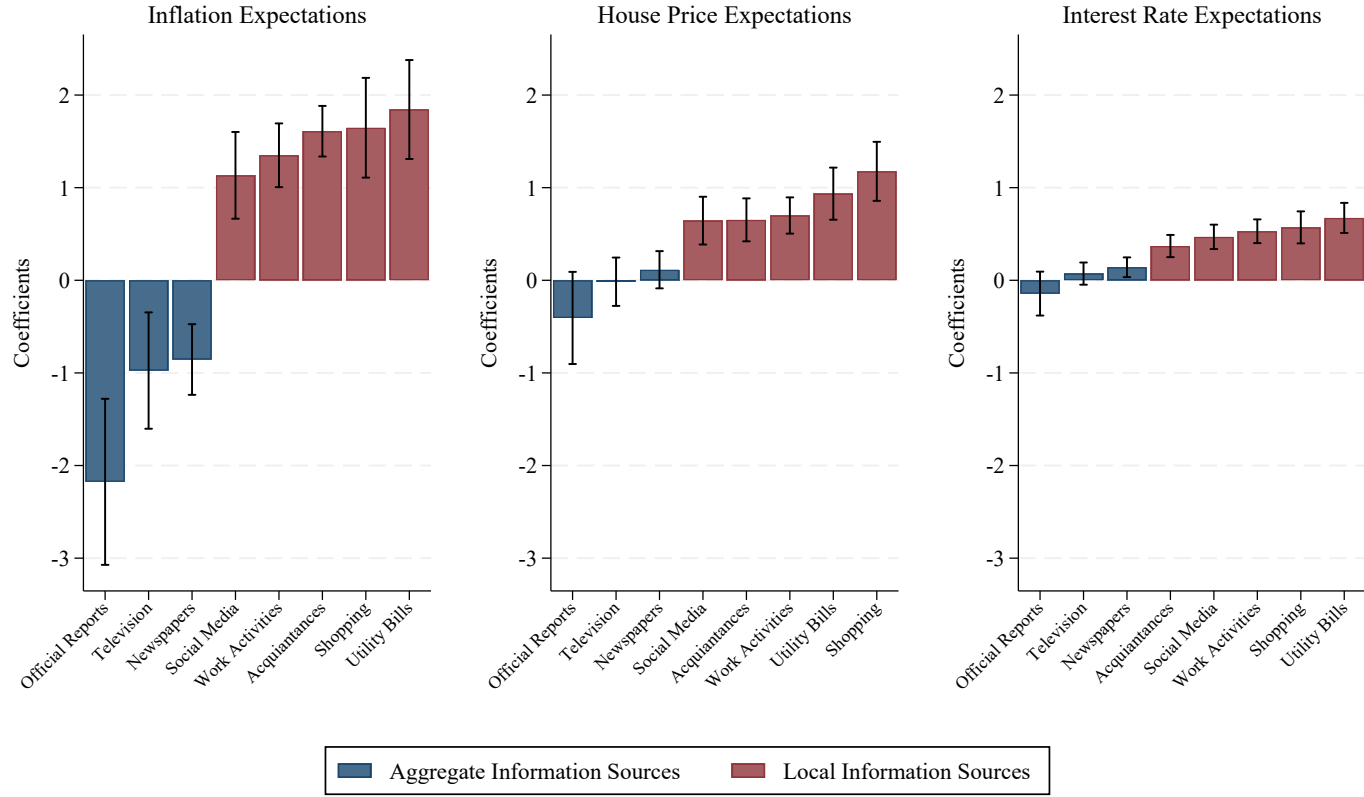


Notes: Each bar in each panel reports an estimated coefficient $\hat{\beta}$ and related 95% confidence intervals from the following specification:

$$Outcome\ Inflation_i = \alpha + \beta Information\ Source_{i,k} + X_i' \delta + \eta_c + \epsilon_i,$$

where $Outcome\ Inflation_i$ is respondent i 's numerical perception for inflation in their country over the 12 months before the survey in the left panel, and the perception error for inflation (defined as the ratio between respondent i 's numerical inflation perception and the actual realized inflation) over the 12 months before the survey in the right panel. Inflation perceptions are based on the survey responses of 46,285 respondents representative of the households across 47 countries in April/May 2023 from the *Global Subjective Expectations Survey*. Each bar refers to the coefficient β estimated in separate regressions run for each information source k at a time, $Information\ Source_k$, which is a dummy that equals 1 if the respondent declares that source k is “Very Important” or “Somewhat Important” for the formation of their subjective expectations, and zero otherwise. Demographic controls X_i include dummy variables for respondents' gender, financial literacy, risk tolerance, numeracy, impulsiveness, patience, expertise with financial decisions, grocery shopping frequency, gas purchase frequency, and ability to cover sudden liquidity shocks; η_c are a full set of country fixed effects (note that for the perception error for inflation, the realized official value is the same for all respondents in the same country). Inflation perception and the perception error for inflation are winsorized at 5-95% level.

Figure 4: Information Sources and Subjective Expectations

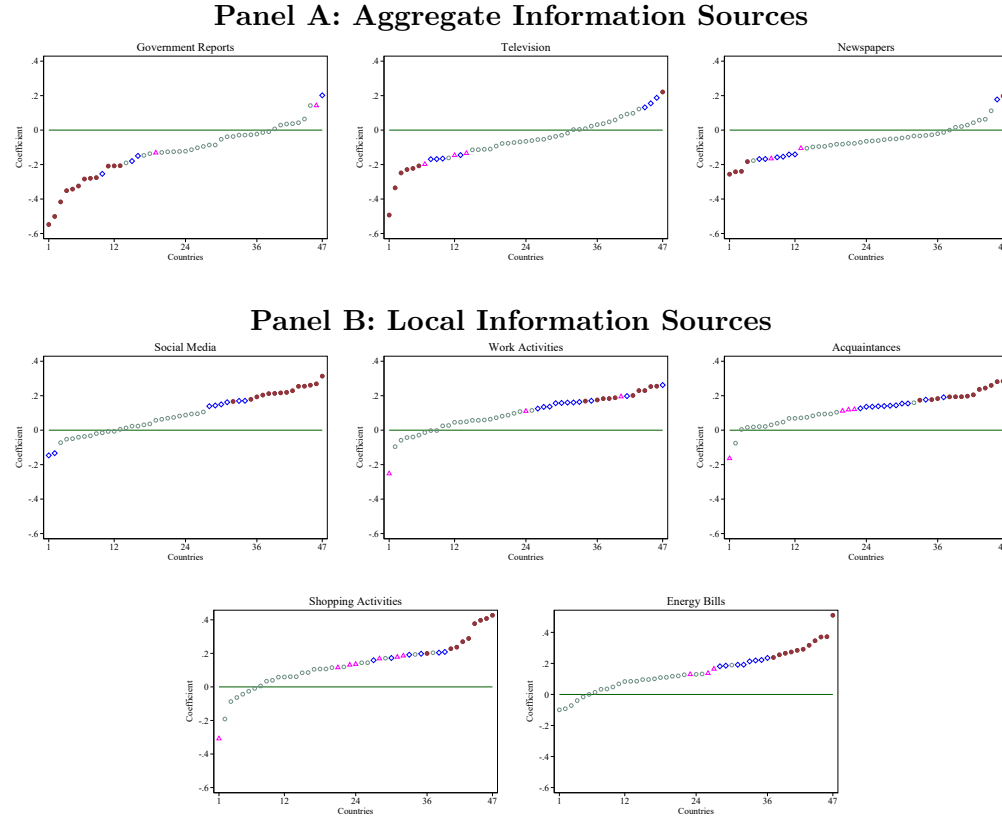


Notes: Each bar in each panel reports an estimated coefficient $\hat{\beta}$ and related 95% confidence intervals from the following specification:

$$Expectations_i = \alpha + \beta Information\ Source_k + X_i' \delta + \eta_c + \epsilon_i,$$

where $Expectations_i$ is respondent i 's numerical expectations for inflation (left panel), house prices (middle panel), and interest rate changes (right panel) in their country over the next 12 months. Expectations are based on the survey responses of 46,285 respondents representative of the households across 47 countries in April/May 2023 from the *Global Subjective Expectations Survey*. Each bar refers to the coefficient β estimated in separate regressions run for each information source k at a time, $Information\ Source_k$, which is a dummy that equals 1 if the respondent declares that source k is "Very Important" or "Somewhat Important" for the formation of their subjective expectations, and zero otherwise. Demographic controls X_i include dummy variables for respondents' gender, financial literacy, risk tolerance, numeracy, impulsiveness, patience, expertise with financial decisions, grocery shopping frequency, gas purchase frequency, and ability to cover sudden liquidity shocks; η_c are a full set of country fixed effects. Expectations are winsorized at 5-95% level.

Figure 5: Subjective Expectations and Information Sources by Country



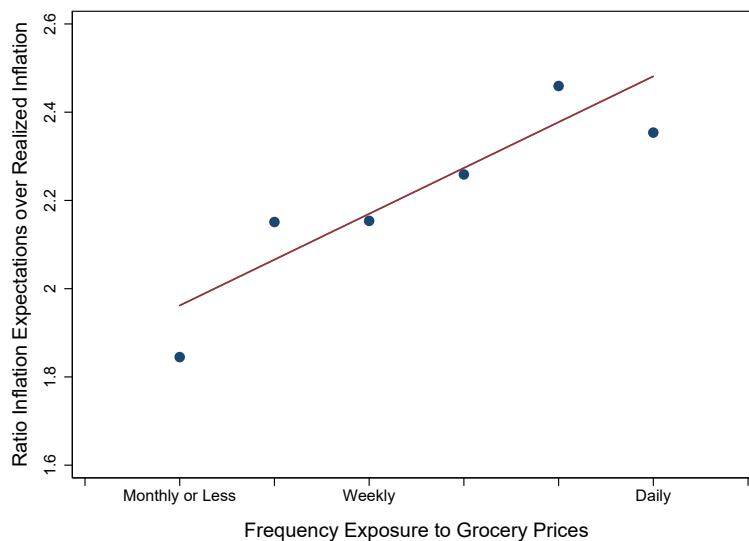
Notes: Each marker reports an estimated coefficient $\hat{\beta}$ from the following specification:

$$Expectations_{i,c} = \alpha + \beta Information\ Source_{i,k} + X_i' \delta + \epsilon_i,$$

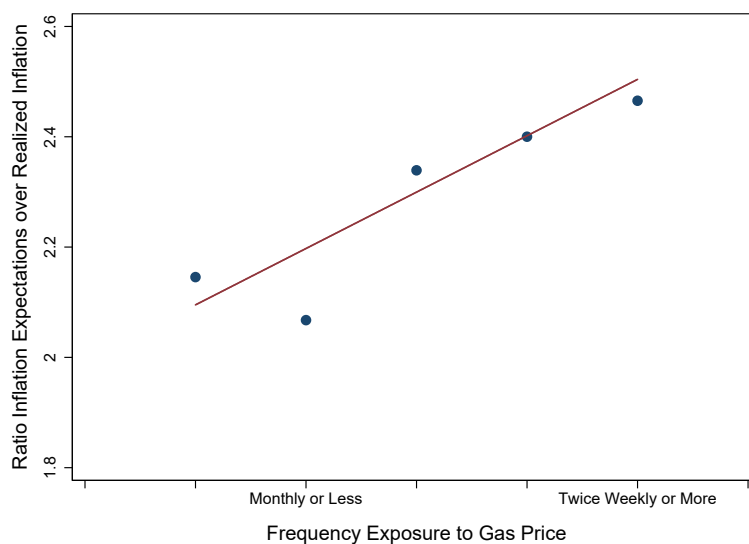
where $Expectations_i$ is respondent i 's numerical expectations for inflation in their country c over the next 12 months. Expectations are standardized at the country level and based on the survey responses of 46,285 respondents representative of the households across 47 countries in April/May 2023 from the *Global Subjective Expectations Survey*. Each marker refers to the coefficient β estimated in separate regressions run for each information source k at a time, $Information\ Source_{i,k}$, which is a dummy that equals 1 if the respondent declares that source k is “Very Important” or “Somewhat Important” for the formation of their subjective expectations, and zero otherwise. Demographic controls X_i include dummy variables for respondents' gender, financial literacy, risk tolerance, numeracy, impulsiveness, patience, expertise with financial decisions, grocery shopping frequency, gas purchase frequency, and ability to cover sudden liquidity shocks. The coefficient estimates for each country-level regression are plotted in order of size from the smallest to the largest. To allow comparison across countries, expectations have been standardized (z-scores) within each country before running each regression. Red circles, blue diamonds, and pink triangles indicate estimates that are statistically significant at the 1%, 5%, and 10% levels, respectively. Expectations are winsorized at 5-95% level.

Figure 6: **Intensity of Exposure to Local Information Sources and Expectation Errors**

Panel A. Frequency of Exposure to Grocery Prices

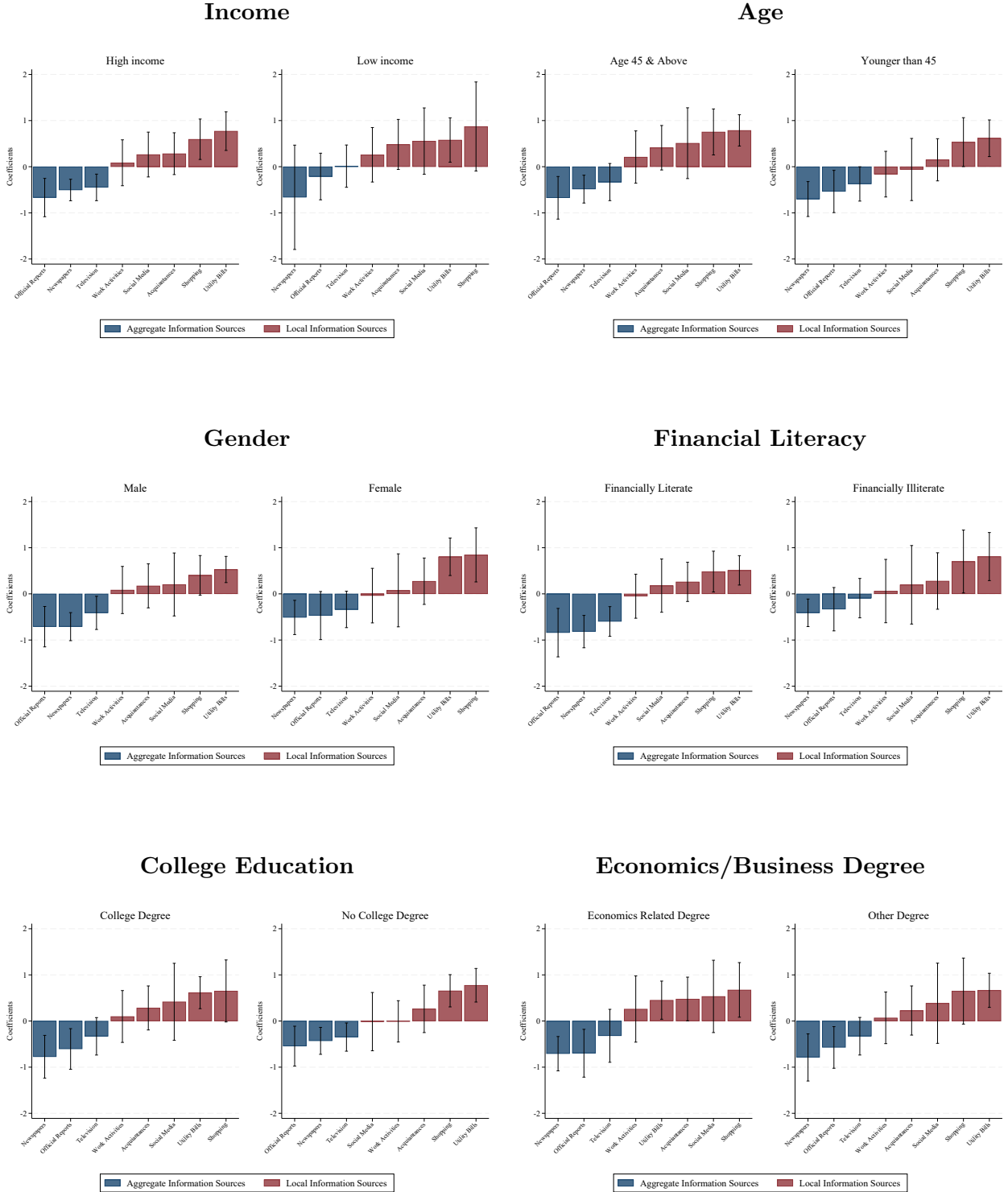


Panel B. Frequency of Exposure to Gas Prices



Notes: The graph reports the average ratio between numerical expectations for inflation in their country over the next 12 months and ex-post realized inflation for each group of survey respondents based on the frequency of exposure to the two local information sources for which we have data—grocery prices via shopping (top panel) and utilities prices via gas station visits (bottom panel). Expectations are based on the survey responses of 46,285 respondents representative of the households across 47 countries in April/May 2023 from the *Global Subjective Expectations Survey*. Ratios are winsorized at 5-95% level.

Figure 7: Information Sources and Subjective Expectations: (Lack of) Variation by Demographics

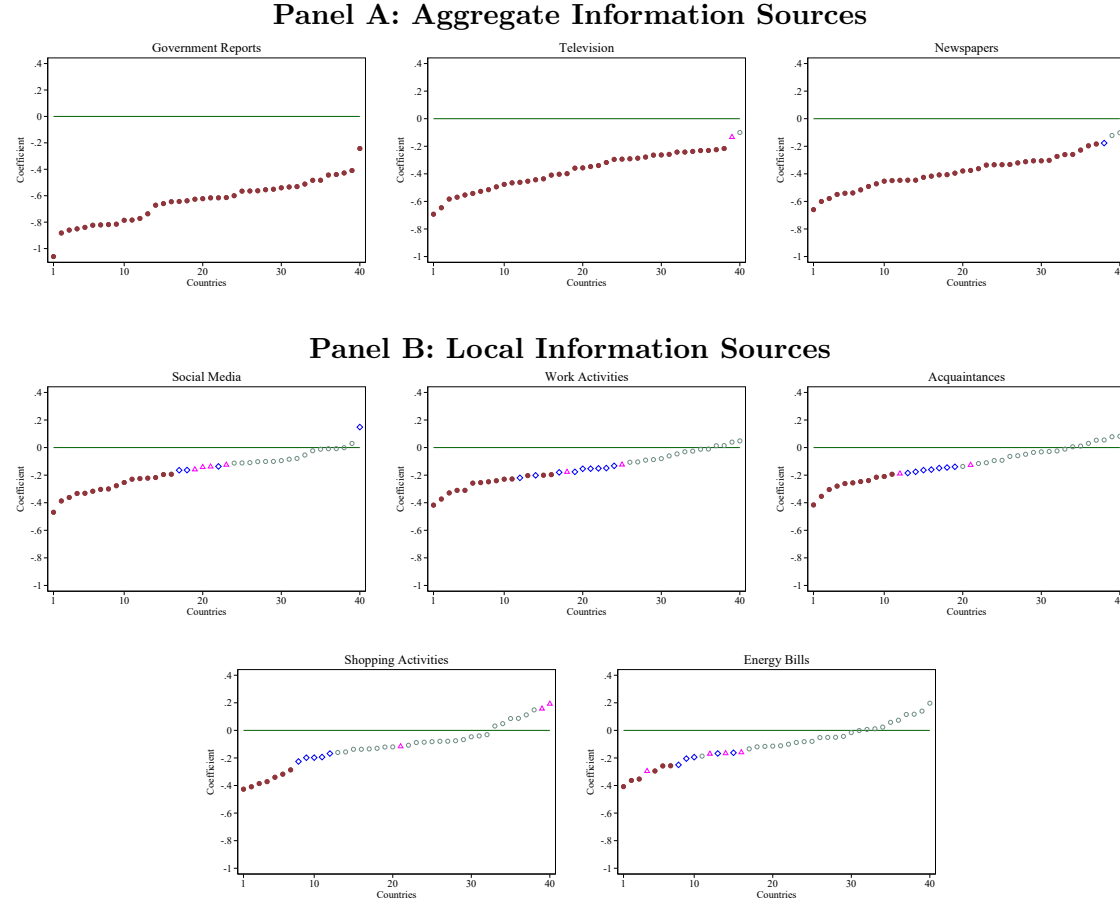


Notes: Each bar in each panel reports an estimated coefficient $\hat{\beta}$ and related 95% confidence intervals from the following specification:

$$\text{Inflation Expectation Error}_i = \alpha + \beta \text{Information Source}_{i,k} + \epsilon_i,$$

where $\text{Inflation Expectation Error}_i$ is respondent i 's expectations error for inflation (defined as the ratio between respondent i 's numerical inflation expectations and the actual realized inflation) over the 12 months after the interview separately by groups of demographic variables. Expectations are based on the survey responses of 46,285 respondents representative of the households across 47 countries in April/May 2023 from the *Global Subjective Expectations Survey*. Each bar refers to the coefficient β estimated in separate regressions run for each information source k at a time, $\text{Information Source}_{i,k}$, which is a dummy that equals 1 if the respondent declares that source k is "Very Important" or "Somewhat Important" for the formation of their subjective expectations, and zero otherwise. These specifications do not include controls to avoid that we absorb systematic variation correlated with the demographic of interest in each Panel, which might mechanically lead us to estimate similar coefficients attached to information sources across sample splits. Expectations are winsorized at 5-95% level.

Figure 8: Distrust in Economic Institutions and Information Sources by Country

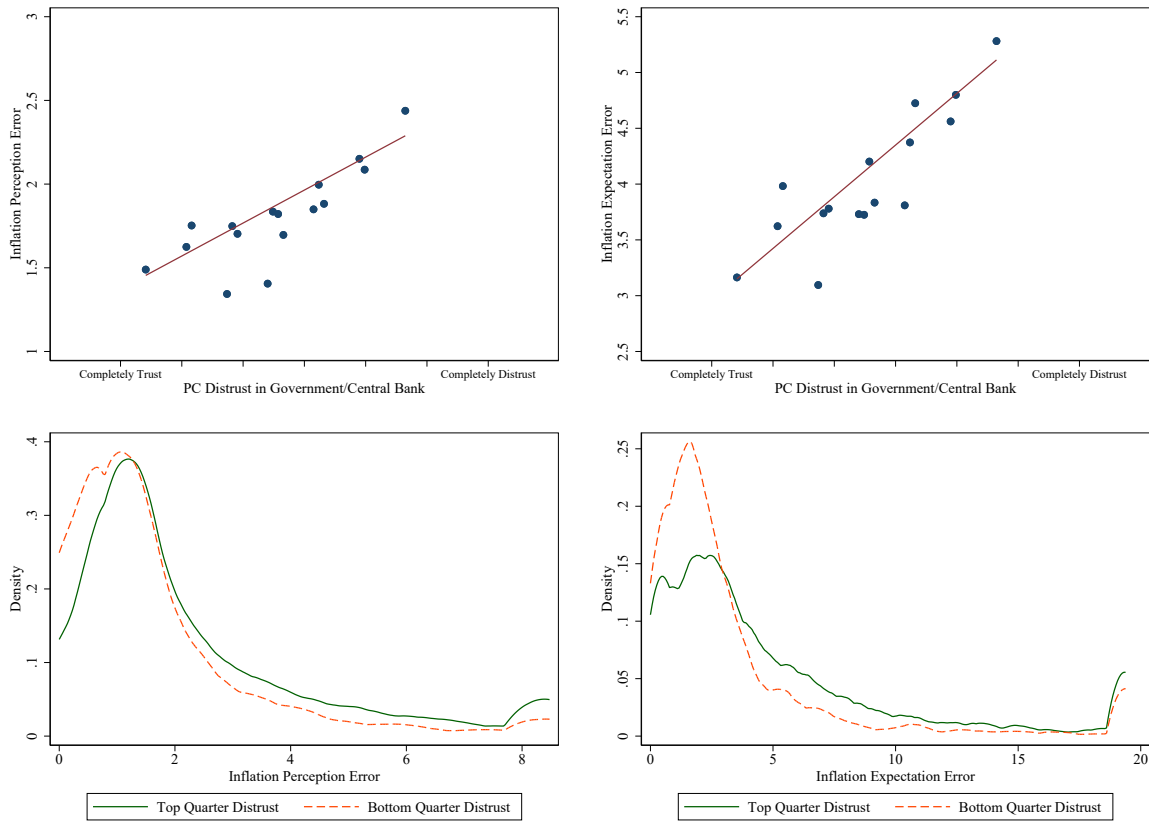


Notes: Each marker reports an estimated coefficient $\hat{\beta}$ from the following specification:

$$Distrust_{i,c} = \alpha + \beta Information\ Source_{i,k} + X_i' \delta + \epsilon_i,$$

where $Distrust_i$ is respondent i 's the principal component of distrust in central bank/government in their country c . Distrust data are based on the survey responses of 36,779 respondents representative of the households across 40 countries in April/May 2023 from the *Global Subjective Expectations Survey*. Each marker refers to the coefficient β estimated in separate regressions run for each information source k at a time, $Information\ Source_k$, which is a dummy that equals 1 if the respondent declares that source k is “Very Important” or “Somewhat Important” for the formation of their subjective expectations, and zero otherwise. Demographic controls X_i include dummy variables for respondents’ gender, financial literacy, risk tolerance, numeracy, impulsiveness, patience, expertise with financial decisions, grocery shopping frequency, gas purchase frequency, and ability to cover sudden liquidity shocks. The coefficient estimates for each country-level regression are plotted in order of size from the smallest to the largest. To allow comparison across countries, trust data have been standardized (z-scores) within each country before running each regression. Red circles, blue diamonds, and pink triangles indicate estimates that are statistically significant at the 1%, 5%, and 10% levels, respectively. Expectations are winsorized at 5-95% level.

Figure 9: Inflation Perception and Expectation Errors by Distrust in Economic Institutions



Notes: The top left (right) plot reports the average ratio between numerical perception (expectations) for inflation in their country over the past (next) 12 months and ex-post realized inflation for each group of survey respondents based on distrust. Distrust is the principal component of distrust in central bank/government in their country. The bottom left (right) plot depicts the histogram of the ratio between numerical perception (expectations) for inflation in their country over the past (next) 12 months and ex-post realized inflation, separately for respondents in the top quarter by distrust (solid-green line) and the bottom quarter by distrust (dashed-orange line). Perceptions and expectations are based on the survey responses of 36,779 respondents representative of the households across 40 countries in April/May 2023 from the *Global Subjective Expectations Survey*. Ratios are winsorized at 5-95% level.

Table 1: Summary Statistics

Variables	Mean	Std. Dev.	p5	p25	p50	p75	p95	Obs.
Panel A. Expectations/Perceptions:								
Inflation Expectations (point estimate)	15.51	17.46	0	5	10	17	70	46,285
Minimum Inflation Expectations	10.40	12.28	0	3	7	10	50	46,284
Maximum Inflation Expectations	21.47	24.78	0	8	12	25	100	46,284
Range of Inflation Expectations	9.89	13.02	0	2	5	10	50	46,284
Inflation Perceptions	14.70	17.20	0	5	10	16.2	70	46,285
Inflation Perception Error	2.02	2.18	0	0.69	1.26	2.55	8.47	46,285
Inflation Expectation Error	4.26	4.91	0	1.25	2.56	5.13	19.38	46,285
Rounded Expectations (multiple of 5)	0.46	0.50	0	0	0	1	1	46,285
House Price Expectations	4.02	8.99	-10	0	0	6	30	46,285
Interest Rate Change Expectations	2.48	5.31	-3	0	0	3	20	46,285
Panel B. Demographic Characteristics:								
Age (18 - 34)	0.39	0.49	0	0	0	1	1	46,285
Age (35 - 44)	0.20	0.40	0	0	0	0	1	46,285
Age (45 - 64)	0.31	0.46	0	0	0	1	1	46,285
Age (65+)	0.10	0.30	0	0	0	0	1	46,285
Female	0.48	0.50	0	0	0	1	1	46,285
Employment	0.65	0.48	0	0	1	1	1	46,285
College Education	0.49	0.50	0	0	0	1	1	46,285
Economics/Business/Finance Degrees	0.12	0.32	0	0	0	0	1	46,285
Married	0.57	0.50	0	0	1	1	1	46,285
Household Size	3.90	1.43	2	3	4	5	6	46,285
High Income	0.60	0.49	0	0	1	1	1	46,285
Panel C. Information Sources:								
Govt. and Official Reports	0.78	0.41	0	1	1	1	1	42,900
TV	0.73	0.44	0	0	1	1	1	43,513
Newspapers	0.68	0.47	0	0	1	1	1	42,064
Social Media	0.63	0.48	0	0	1	1	1	43,203
Work Activities	0.77	0.42	0	1	1	1	1	40,257
Acquaintances	0.74	0.44	0	0	1	1	1	43,400
Shopping	0.88	0.32	0	1	1	1	1	43,826
Utility Bills	0.89	0.31	0	1	1	1	1	43,642
Panel D. Distrust in Economic Institutions:								
Principal Component Distrust in Govt./Central Bank	0.00	1.22	-1.43	-1.43	-0.01	1.42	1.42	36,779

Notes: This table summarizes respondents' expectations, perceptions, and realized inflation in Panel A, demographic characteristics in Panel B, importance of information sources in Panel C, and the principal component of trust in Government/Central bank in Panel D. Data are from the survey responses of 46,285 respondents representative of the households across 47 countries in April/May 2023 from the *Global Subjective Expectations Survey*. Continuous variables are winsorized at 5-95% level.

Table 2: Information Sources, Perceptions, and Expectations

	(1)	(2)	(3)	(4)	(5)	(6)
	Perception Error			Expectation Error		
	$\frac{Inflation\ Perceptions_t}{Realized\ Inflation_t}$			$\frac{Inflation\ Expectations_t}{Realized\ Inflation_{t+1}}$		
<u>Aggregate Sources</u>						
Govt. & Official Reports	-0.33*** (0.10)	-0.31*** (0.09)	-0.28*** (0.04)	-0.57** (0.21)	-0.46** (0.19)	-0.50*** (0.07)
Newspapers	-0.15** (0.07)	-0.18** (0.08)	-0.04 (0.03)	-0.58** (0.24)	-0.54* (0.27)	-0.18** (0.07)
TV	-0.10 (0.08)	-0.10 (0.08)	-0.08** (0.04)	-0.20 (0.13)	-0.23* (0.13)	-0.22*** (0.07)
<u>Local Sources</u>						
Shopping	0.43*** (0.10)	0.37*** (0.08)	0.32*** (0.04)	0.62*** (0.20)	0.65*** (0.20)	0.41*** (0.07)
Utility Bills	0.31*** (0.07)	0.27*** (0.05)	0.29*** (0.05)	0.76*** (0.17)	0.71*** (0.15)	0.58*** (0.10)
Social Media	0.04 (0.09)	0.12 (0.08)	0.06** (0.03)	0.25 (0.31)	0.33 (0.25)	0.23*** (0.05)
Friends	0.07 (0.05)	0.09* (0.05)	0.10*** (0.02)	0.17 (0.17)	0.15 (0.15)	0.26*** (0.04)
Work Activities	0.10 (0.07)	0.08 (0.05)	0.09*** (0.03)	-0.15 (0.18)	-0.056 (0.12)	0.17*** (0.05)
Demographic Controls		X	X		X	X
Country FE			X			X
R^2	0.02	0.03	0.23	0.01	0.03	0.33
Observations	46,285	46,285	46,285	46,285	46,285	46,285

Notes: We report estimates from the following specification:

$$Outcome\ Inflation_i = \alpha + \sum_k \beta_k Information\ Source_{i,k} + X_i' \delta + \eta_c + \epsilon_i,$$

where *Outcome Inflation_i* is respondent *i*'s perception error for inflation (defined as the ratio between respondent *i*'s numerical inflation perception and the actual realized inflation) over the 12 months before the interview in columns (1)–(3) and the expectations error for inflation (defined as the ratio between respondent *i*'s numerical inflation expectations and the actual realized inflation) over the 12 months after the interview in columns (4)–(6). Inflation perceptions and expectations data are based on the survey responses of 46,285 respondents representative of the households across 47 countries in April/May 2023 from the *Global Subjective Expectations Survey*. *Information Source_k* is a dummy that equals 1 if the respondent declares that source *k* is “Very Important” or “Somewhat Important” for the formation of their subjective expectations, and zero otherwise. Demographic controls *X_i* include dummy variables for respondents' gender, financial literacy, risk tolerance, numeracy, impulsiveness, patience, expertise with financial decisions, grocery shopping frequency, gas purchase frequency, and ability to cover sudden liquidity shocks; η_c are a full set of country fixed effects (note that realized inflation is the same for all respondents in the same country). Inflation perception and the perception error for inflation are winsorized at 5-95% level. Standard errors clustered at the country level are reported in parentheses. *p<0.10, **p<0.05, ***p<0.01

Table 3: Distrust, Perceptions, and Expectations

	(1)	(2)	(3)	(4)	(5)	(6)
	Perception Error			Expectation Error		
	$\frac{Inflation\ Perceptions_t}{Realized\ Inflation_t}$			$\frac{Inflation\ Expectations_t}{Realized\ Inflation_{t+1}}$		
Distrust in Economic Institutions	0.20*** (0.05)	0.14*** (0.04)	0.16*** (0.02)	0.46*** (0.15)	0.36*** (0.11)	0.36*** (0.04)
Distrust in Corporations		0.03 (0.03)	0.04*** (0.01)		-0.04 (0.11)	0.04 (0.03)
Distrust in the Stock Market		0.08 (0.05)	0.04** (0.02)		0.02 (0.13)	0.05 (0.04)
Preferred Party in Power		0.05 (0.05)	-0.01 (0.03)		-0.10 (0.13)	-0.02 (0.07)
Demographic Controls		X	X		X	X
Country FE			X			X
R^2	0.02	0.03	0.19	0.01	0.04	0.36
Observations	36,779	34,785	34,785	36,779	34,785	34,785

Notes: We report estimates from the following specification:

$$Outcome\ Inflation_i = \alpha + \sum_l \beta_l distrust_{i,l} + \gamma Preferred\ Party_i + X_i' \delta + \eta_c + \epsilon_i,$$

where $Outcome\ Inflation_i$ is respondent i 's perception error for inflation (defined as the ratio between respondent i 's numerical inflation perception and the actual realized inflation) over the 12 months before the interview in columns (1)–(3) and the expectations error for inflation (defined as the ratio between respondent i 's numerical inflation expectations and the actual realized inflation) over the 12 months after the interview in columns (4)–(6). Inflation perceptions and expectations data are based on the survey responses of 46,285 respondents representative of the households across 47 countries in April/May 2023 from the *Global Subjective Expectations Survey*. $distrust_l$ is the principal component of trust in central bank/government in their country or distrust towards private corporations or the stock market. *Preferred Party* is a dummy variable which equals 1 if respondents indicate that their preferred political party is in power/part of the majority coalition in their country. Demographic controls X_i include dummy variables for respondents' gender, financial literacy, risk tolerance, numeracy, impulsiveness, patience, expertise with financial decisions, grocery shopping frequency, gas purchase frequency, and ability to cover sudden liquidity shocks; η_c are a full set of country fixed effects (note that realized inflation is the same for all respondents in the same country). Inflation perception and the perception error for inflation are winsorized at 5-95% level. Standard errors clustered at the country level are reported in parentheses. *p<0.10, **p<0.05, ***p<0.01

Table 4: **Past Economic Conditions and Distrust in Economic Institutions**

	(1)	(2)	(3)	(4)
	Distrust in Economic Institutions			
10-year Average Inflation	0.03 (0.03)		-0.02 (0.03)	
10-year St. Dev. Inflation		0.17*** (0.04)	0.20*** (0.05)	
10-year Coeff. Variation Inflation				0.16** (0.06)
Demographic Controls	X	X	X	X
R^2	0.10	0.13	0.13	0.11
Observations	35,047	35,047	35,047	35,047

Notes: We report estimates from the following specification:

$$Distrust_{i,c} = \alpha + \beta Inflation_c + X_i' \delta + \epsilon_i,$$

where $Distrust_i$ is respondent i 's the principal component of distrust in central bank/government in their country c . Distrust data are based on the survey responses of 46,285 respondents representative of the households across 47 countries in April/May 2023 from the *Global Subjective Expectations Survey*. $Inflation_c$ is the 10-year average realized year on year inflation rate at the monthly frequency in country (c) before our survey (columns (1) and (3)), the 10-year standard deviation (columns (2) and (3)), and on the 10-year coefficient of variation (column (4)). Demographic controls X_i include dummy variables for respondents' gender, financial literacy, risk tolerance, numeracy, impulsiveness, patience, expertise with financial decisions, grocery shopping frequency, gas purchase frequency, and ability to cover sudden liquidity shocks. Standard errors clustered at the country level are reported in parentheses. *p<0.10, **p<0.05, ***p<0.01

Online Appendix:
Information and Macroeconomic Expectations:
Global Evidence

Francesco D'Acunto and Michael Weber

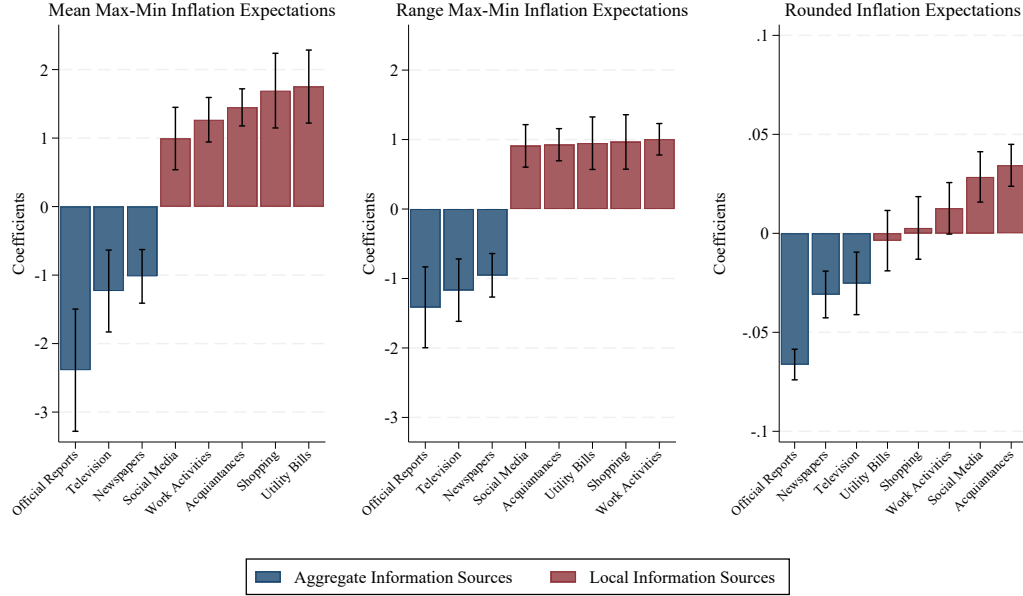
Not for Publication

Table A.1: **Summary Statistics: Demographic Variables by Country**

	AR	AT	AU	BD	BR	CA	CL	CN	CO	DE	EG	FI	FR	GR	HR	HU
Age: 18-34	0.37	0.26	0.27	0.79	0.44	0.26	0.35	0.51	0.53	0.25	0.76	0.27	0.25	0.24	0.26	0.25
Age: 35-44	0.22	0.16	0.22	0.15	0.22	0.19	0.18	0.19	0.24	0.21	0.15	0.19	0.21	0.19	0.18	0.18
Age: 45-64	0.34	0.35	0.33	0.06	0.29	0.33	0.41	0.26	0.22	0.43	0.09	0.39	0.42	0.36	0.35	0.33
Age: 65+	0.08	0.23	0.17	0.01	0.04	0.21	0.06	0.04	0.01	0.11	0.00	0.15	0.11	0.20	0.21	0.24
College Education	0.40	0.21	0.46	0.52	0.51	0.86	0.68	0.50	0.51	0.32	0.76	0.40	0.25	0.52	0.46	0.22
Female	0.52	0.51	0.50	0.24	0.51	0.52	0.51	0.46	0.51	0.53	0.39	0.51	0.55	0.48	0.50	0.53
Financial Literacy	0.75	0.59	0.48	0.45	0.55	0.60	0.63	0.19	0.58	0.55	0.35	0.52	0.54	0.55	0.69	0.75
Full-time Employment	0.76	0.51	0.57	0.44	0.81	0.64	0.69	0.86	0.72	0.60	0.75	0.60	0.62	0.60	0.48	0.48
Married	0.56	0.50	0.55	0.43	0.62	0.53	0.50	0.72	0.55	0.50	0.62	0.54	0.60	0.63	0.62	0.59
	ID	IE	IL	IN	IT	JP	KE	MA	MX	MY	NG	NL	NO	NZ	PH	PK
Age: 18-34	0.63	0.28	0.35	0.78	0.25	0.21	0.57	0.43	0.49	0.45	0.60	0.29	0.30	0.34	0.61	0.64
Age: 35-44	0.21	0.21	0.19	0.09	0.19	0.17	0.24	0.31	0.21	0.20	0.21	0.17	0.17	0.14	0.21	0.19
Age: 45-64	0.15	0.33	0.39	0.12	0.45	0.50	0.13	0.23	0.26	0.33	0.18	0.34	0.33	0.33	0.17	0.16
Age: 65+	0.00	0.19	0.07	0.01	0.11	0.11	0.06	0.02	0.04	0.02	0.01	0.20	0.20	0.19	0.01	0.01
College Education	0.49	0.42	0.53	0.64	0.22	0.47	0.44	0.53	0.77	0.55	0.73	0.36	0.45	0.31	0.60	0.42
Female	0.47	0.51	0.51	0.33	0.51	0.48	0.51	0.51	0.53	0.48	0.32	0.49	0.49	0.52	0.55	0.35
Financial Literacy	0.23	0.47	0.51	0.25	0.43	0.38	0.41	0.37	0.51	0.48	0.41	0.59	0.51	0.58	0.38	0.38
Full-time Employment	0.70	0.60	0.74	0.73	0.61	0.65	0.71	0.61	0.76	0.73	0.80	0.60	0.55	0.60	0.68	0.52
Married	0.53	0.62	0.58	0.48	0.63	0.59	0.53	0.50	0.61	0.53	0.47	0.60	0.55	0.60	0.49	0.56
	AE	CH	ES	GB	KR	PL	RU	SA	SE	SG	TH	TR	US	ZA	ZW	Total
Age: 18-34	0.27	0.27	0.26	0.27	0.33	0.27	0.37	0.47	0.30	0.29	0.35	0.37	0.27	0.50	0.60	0.34
Age: 35-44	0.38	0.17	0.22	0.22	0.24	0.20	0.18	0.23	0.19	0.17	0.23	0.21	0.19	0.32	0.22	0.22
Age: 45-64	0.34	0.34	0.45	0.42	0.34	0.32	0.34	0.29	0.34	0.41	0.39	0.42	0.31	0.15	0.17	0.34
Age: 65+	0.01	0.21	0.06	0.10	0.09	0.22	0.10	0.00	0.17	0.13	0.02	0.01	0.23	0.02	0.00	0.10
College Education	0.81	0.27	0.38	0.43	0.67	0.36	0.49	0.61	0.21	0.53	0.69	0.66	0.37	0.34	0.61	0.49
Female	0.24	0.49	0.50	0.54	0.47	0.52	0.51	0.39	0.49	0.51	0.51	0.50	0.52	0.51	0.50	0.48
Financial Literacy	0.35	0.44	0.54	0.52	0.65	0.62	0.62	0.32	0.50	0.52	0.32	0.67	0.44	0.47	0.51	0.50
Full-time Employment	0.89	0.62	0.69	0.61	0.68	0.52	0.57	0.67	0.60	0.78	0.79	0.70	0.46	0.69	0.65	0.66
Married	0.81	0.51	0.59	0.57	0.55	0.65	0.62	0.69	0.52	0.60	0.50	0.70	0.49	0.46	0.41	0.58

Notes: This table summarizes respondents' demographic characteristics at the country level. Data are from the survey responses of 46,285 respondents representative of the households across 47 countries in April/May 2023 from the *Global Subjective Expectations Survey*.

Figure A.1: Range and Mean Inflation Expectations by Information Sources

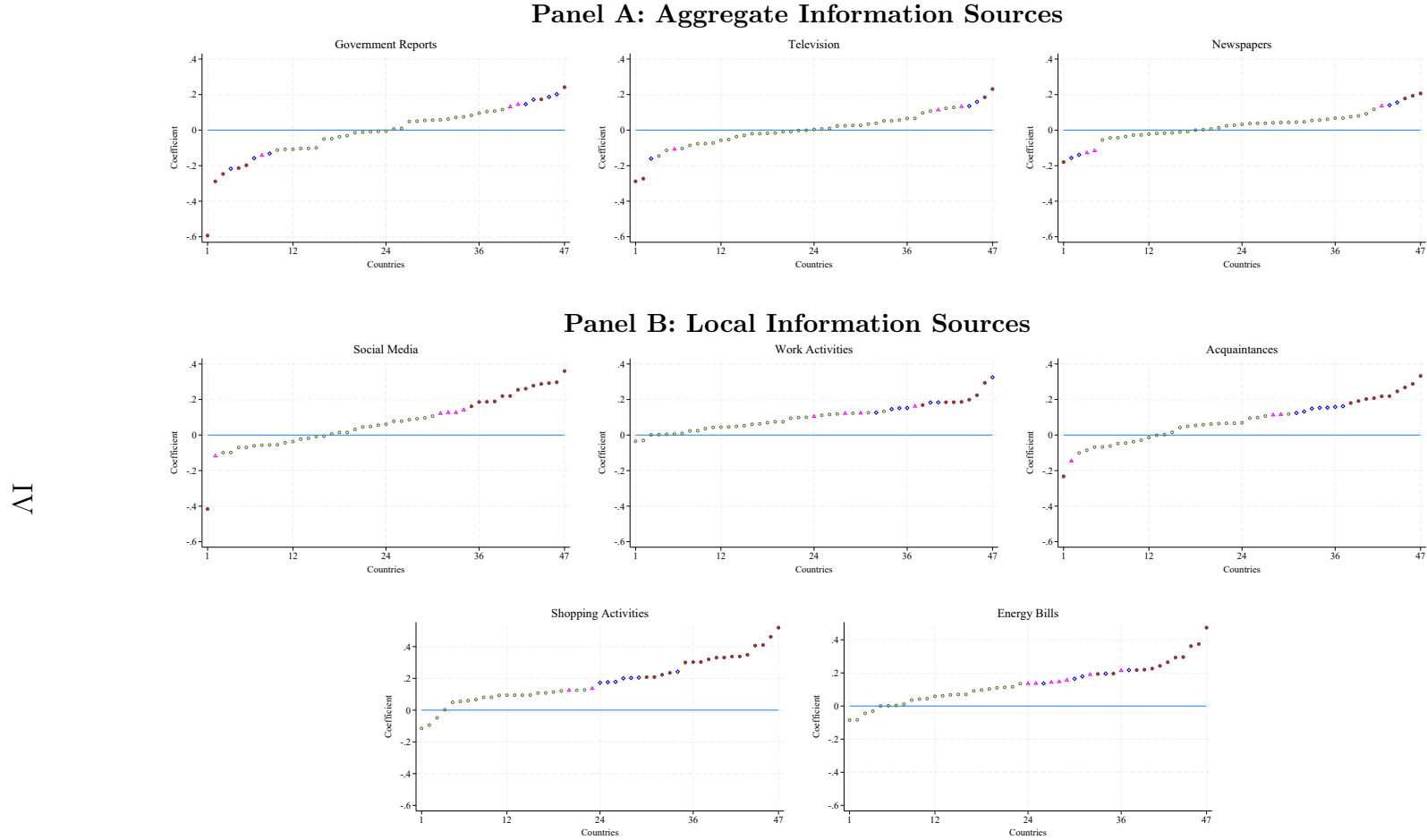


Notes: Each bar in each panel reports an estimated coefficient $\hat{\beta}$ and related 95% confidence intervals from the following specification:

$$Outcome\ Inflation_i = \alpha + \beta Information\ Source_{i,k} + X_i' \delta + \eta_c + \epsilon_i,$$

where $Outcome\ Inflation_i$ is the average of respondent i 's minimum and maximum possible expected inflation in their country over the 12 months after the survey in the left panel; the range (maximum minus minimum) in the middle panel; and, a dummy variable that equals 1 if respondent i reports a point estimate for inflation expectations that is a multiple of 5 (rounded expectations). Inflation expectations are based on the survey responses of 46,285 respondents representative of the households across 47 countries in April/May 2023 from the *Global Subjective Expectations Survey*. Each bar refers to the coefficient β estimated in separate regressions run for each information source k at a time, $Information\ Source_k$, which is a dummy that equals 1 if the respondent declares that source k is "Very Important" or "Somewhat Important" for the formation of their subjective expectations, and zero otherwise. Demographic controls X_i include dummy variables for respondents' gender, financial literacy, risk tolerance, numeracy, impulsiveness, patience, expertise with financial decisions, grocery shopping frequency, gas purchase frequency, and ability to cover sudden liquidity shocks; η_c are a full set of country fixed effects (note that for the perception error for inflation, the realized official value is the same for all respondents in the same country). Inflation perception and the perception error for inflation are winsorized at 5-95% level.

Figure A.2: Subjective House Price Expectations and Information Sources by Country



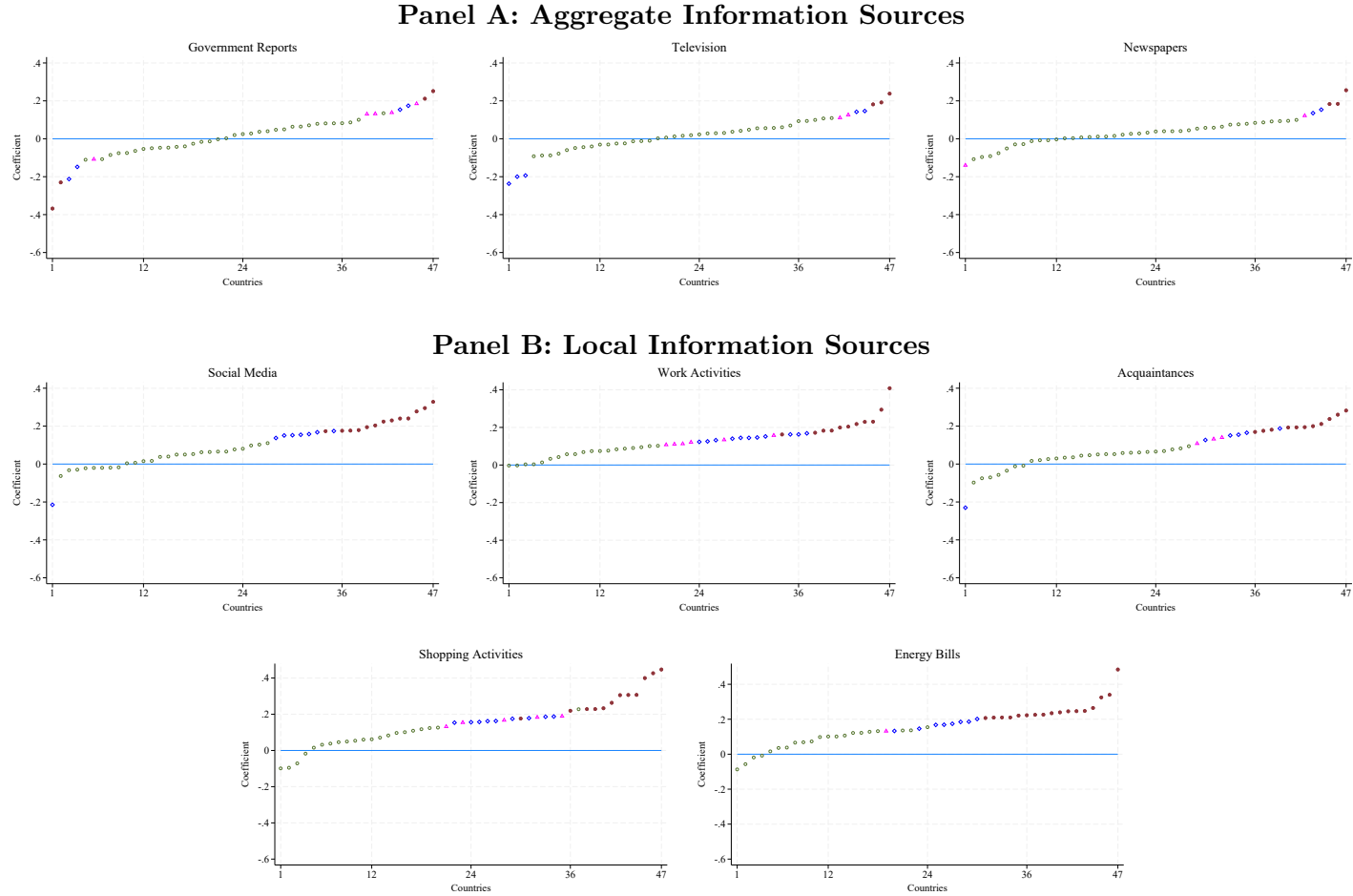
Notes: Each marker reports an estimated coefficient $\hat{\beta}$ from the following specification:

$$Expectations_{i,c} = \alpha + \beta Information\ Source_{i,k} + X'_i \delta + \epsilon_i,$$

where $Expectations_i$ is respondent i 's numerical expectations for house prices in their country c over the next 12 months. Expectations are standardized at the country level and based on the survey responses of 46,285 respondents representative of the households across 47 countries in April/May 2023 from the *Global Subjective Expectations Survey*. Each marker refers to the coefficient β estimated in separate regressions run for each information source k at a time, $Information\ Source_k$, which is a dummy that equals 1 if the respondent declares that source k is “Very Important” or “Somewhat Important” for the formation of their subjective expectations, and zero otherwise. Demographic controls X_i include dummy variables for respondents' gender, financial literacy, risk tolerance, numeracy, impulsiveness, patience, expertise with financial decisions, grocery shopping frequency, gas purchase frequency, and ability to cover sudden liquidity shocks. The coefficient estimates for each country-level regression are plotted in order of size from the smallest to the largest. To allow comparison across countries, expectations have been standardized (z-scores) within each country before running each regression. Red circles, blue diamonds, and pink triangles indicate estimates that are statistically significant at the 1%, 5%, and 10% levels, respectively. Expectations are winsorized at 5-95% level.

Figure A.3: Subjective Interest Rate Change Expectations and Information Sources by Country

Λ



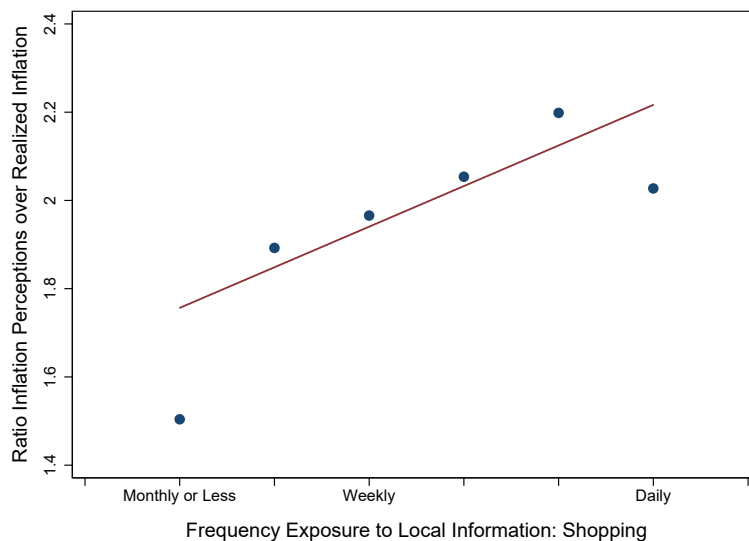
Notes: Each marker reports an estimated coefficient $\hat{\beta}$ from the following specification:

$$Expectations_{i,c} = \alpha + \beta Information\ Source_{i,k} + X'_i \delta + \epsilon_i,$$

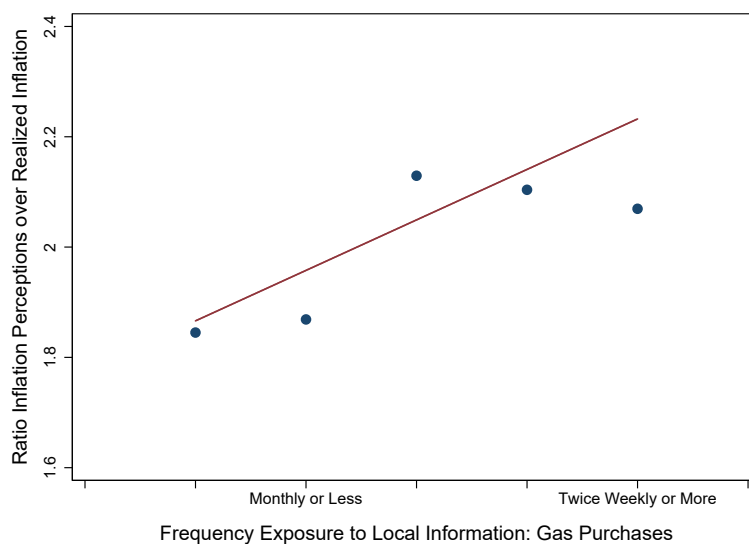
where $Expectations_i$ is respondent i 's numerical expectations for interest rate changes in their country c over the next 12 months. Expectations are standardized at the country level and based on the survey responses of 46,285 respondents representative of the households across 47 countries in April/May 2023 from the *Global Subjective Expectations Survey*. Each marker refers to the coefficient β estimated in separate regressions run for each information source k at a time, $Information\ Source_k$, which is a dummy that equals 1 if the respondent declares that source k is “Very Important” or “Somewhat Important” for the formation of their subjective expectations, and zero otherwise. Demographic controls X_i include dummy variables for respondents' gender, financial literacy, risk tolerance, numeracy, impulsiveness, patience, expertise with financial decisions, grocery shopping frequency, gas purchase frequency, and ability to cover sudden liquidity shocks. The coefficient estimates for each country-level regression are plotted in order of size from the smallest to the largest. To allow comparison across countries, expectations have been standardized (z-scores) within each country before plotting each regression. Red circles, blue diamonds, and pink triangles indicate estimates that are statistically significant at the 1%, 5%, and 10% levels, respectively. Expectations are winsorized at 5-95% level.

Figure A.4: Intensity of Exposure to Local Information Sources and Perception Errors

Panel A. Frequency of Exposure to Grocery Prices

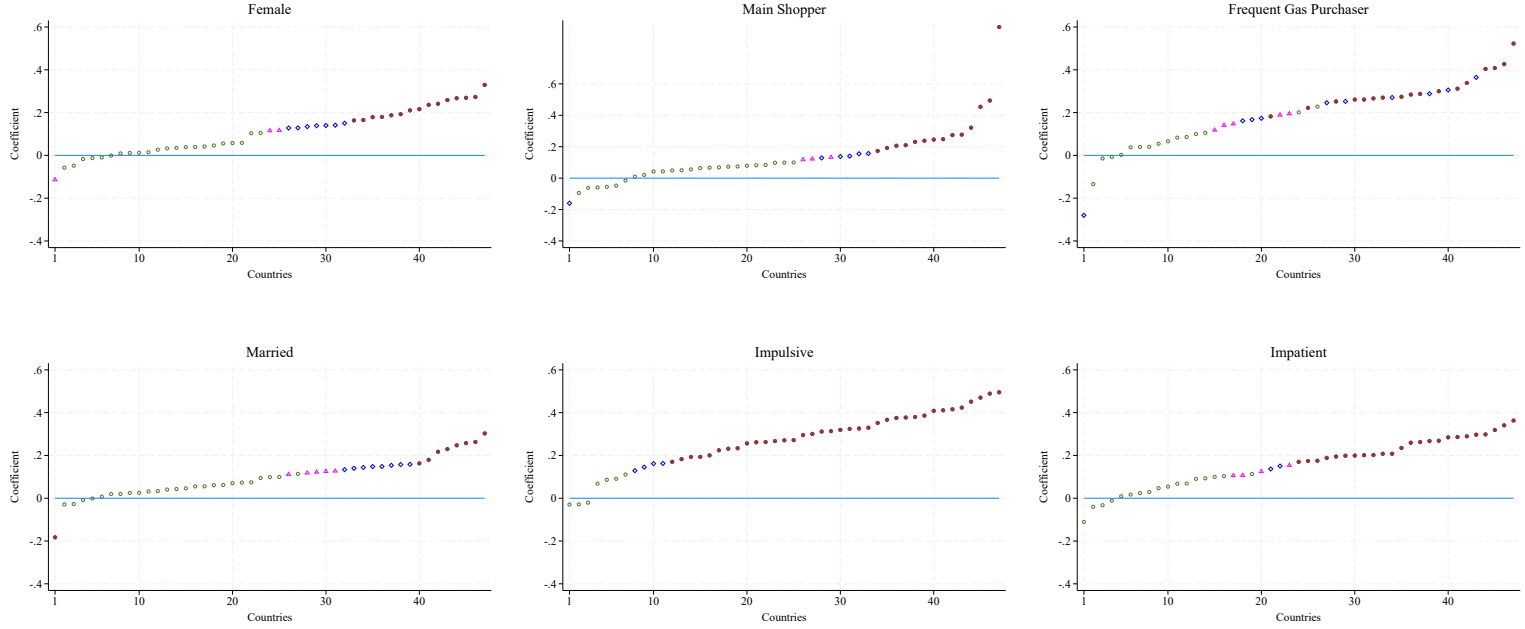


Panel B. Frequency of Exposure to Gas Prices



Notes: The graph reports the average ratio between numerical perceptions for inflation in their country over the 12 months before the survey and the actual realized inflation rate for each group of survey respondents based on the frequency of exposure to the two local information sources for which we have data—grocery prices via shopping (top panel) and utilities prices via gas station visits (bottom panel). Perceptions are based on the survey responses of 46,285 respondents representative of the households across 47 countries in April/May 2023 from the *Global Subjective Expectations Survey*. Ratios are winsorized at 5-95% level.

Figure A.5: Sorting into Local Information Sources

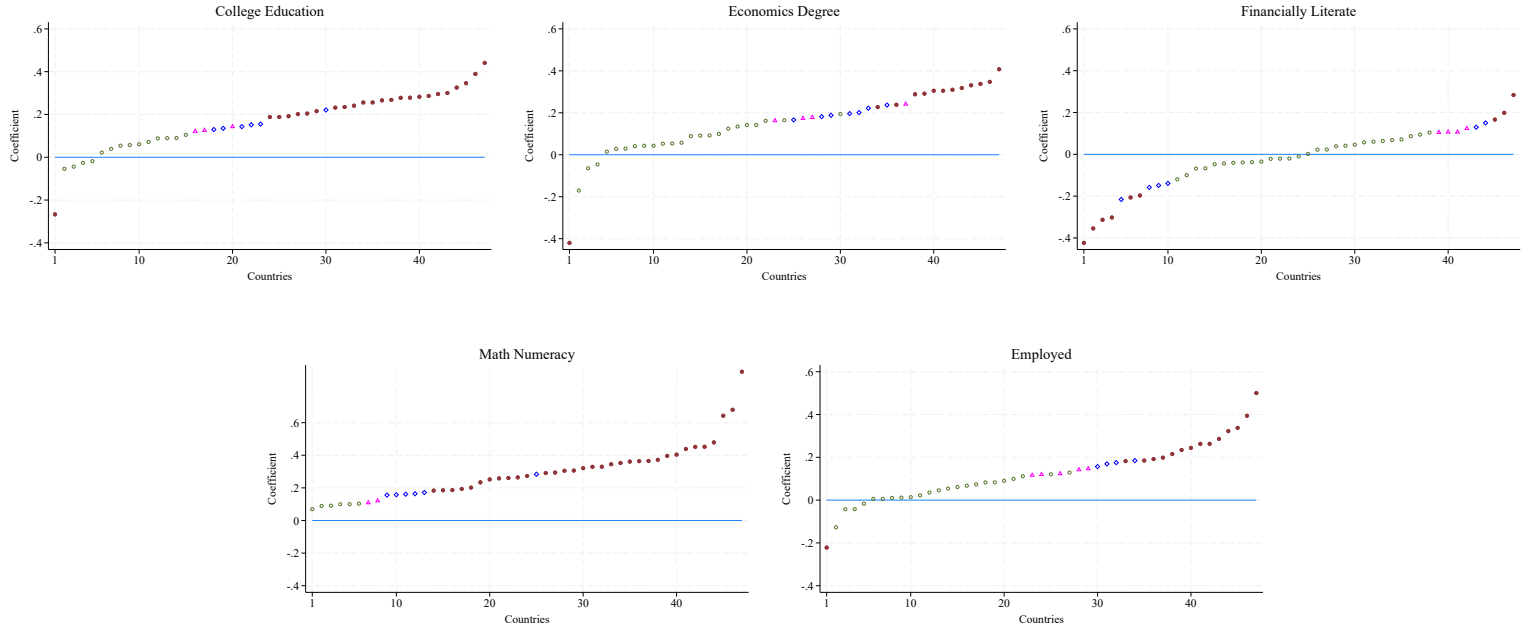


Notes: Each marker reports an estimated coefficient $\hat{\beta}$ from the following specification:

$$Local\ Sources_{i,k,c} = \alpha + \beta X'_i + \epsilon_i,$$

where $Local\ Sources_{i,k,c}$ is the sum of dummy variables that equal 1 if the respondent i declares that each local information source is “Very Important” or “Somewhat Important” for the formation of their subjective expectations, and zero otherwise, divided by the number of local information sources. This variable thus varies between 0 (no local source is important) to 1 (all five local sources are important). Data are based on the survey responses of 46,285 respondents representative of the households across 47 countries in April/May 2023 from the *Global Subjective Expectations Survey*. Each marker refers to the coefficient β estimated in separate regressions run for each demographic variable as indicated on the top of the graph. The coefficient estimates for each country-level regression are plotted in order of size from the smallest to the largest. To allow comparison across countries c , $Source$ have been standardized (z-scores) within each country before running each regression. Red circles, blue diamonds, and pink triangles indicate estimates that are statistically significant at the 1%, 5%, and 10% levels, respectively. Expectations are winsorized at 5-95% level.

Figure A.6: Sorting into Aggregate Information Sources



Notes: Each marker reports an estimated coefficient $\hat{\beta}$ from the following specification:

$$Aggregate\ Sources_{i,k,c} = \alpha + \beta X_i' + \epsilon_i,$$

where $Aggregate\ Sources_{i,c}$ is the sum of dummy variables that equal 1 if the respondent i declares that each aggregate information source is “Very Important” or “Somewhat Important” for the formation of their subjective expectations, and zero otherwise, divided by the number of aggregate information sources. This variable thus varies between 0 (no local source is important) to 1 (all three local sources are important). Data are based on the survey responses of 46,285 respondents representative of the households across 47 countries in April/May 2023 from the *Global Subjective Expectations Survey*. Each marker refers to the coefficient β estimated in separate regressions run for each demographic variable as indicated on the top of the graph. The coefficient estimates for each country-level regression are plotted in order of size from the smallest to the largest. To allow comparison across countries c , $Source$ have been standardized (z-scores) within each country before running each regression. Red circles, blue diamonds, and pink triangles indicate estimates that are statistically significant at the 1%, 5%, and 10% levels, respectively. Expectations are winsorized at 5-95% level.

1 Survey Questions

Below we report the original survey questions for the United States with answer choices in English.

Question 1 *What is your age?*

Select from a dropbox

Question 2 *How would you describe your present employment situation?*

- Job in the private sector
- Job in government
- Self-Employed
- Homemaker
- Student
- Retired
- Unemployed
- Prefer not to answer

Question 3 *What is your current marital status?*

- Single
- Never married
- Married
- Separated
- Divorced
- Widowed
- Living with a partner
- Don't know/No opinion

Question 4 *Which category represents the total combined income of all members of your HOUSEHOLD during the past 12 months?*

This includes money from jobs, net income from business, farm or rent, pensions, dividends, interest, social security payments and any other money income received by members of your family who are 15 years of age or older.

- Under 20 thousand dollars
- 20 to under 35 thousand
- 35 to under 50 thousand
- 50 to under 75 thousand
- 75 to under 100 thousand
- 100 thousand or more

If response “100 thousand or more” in Question 4 is selected

Question 4 *Is that ...*

- 100 to under 150 thousand
- 150 to under 200 thousand
- 200 to under 250 thousand
- 250 thousand or more

Question 5 *What is the last grade or class you completed in school?*

- None, or grade 1-8
- High school incomplete (grades 9-11)
- High school diploma or equivalent, no further schooling
- Technical or vocational school after high school
- Some college, no degree
- Associate’s or two-year college degree
- Four-year college degree
- Graduate or professional school after college, no degree
- Graduate or professional degree

[ASK ONLY OF THOSE WHO ATTENDED COLLEGE IN PREVIOUS QUESTION]

Question 6 *What was your primary area of study in college?*

[RANDOMIZE]

- Economics/Business
- Social Sciences or Law
- Engineering or Sciences (Math, Physics, Biology, Chemistry, etc.)
- Medical or paramedical
- Humanities
- Other [ANCHOR]

Question 7 *Please tell us how many of the following people usually live in your primary residence other than yourself (including those who are temporarily away).*

- 1 person
- 2 people
- 3 people
- 4 people
- 5 or more people
- None

Question 8 *In the next 12 months do you expect your household income to increase, decrease, or stay about the same?*

- Increase
- Decrease
- Stay about the same
- Don't know/prefer not to answer [ANCHOR]

Question 9 *Suppose that you had to make an unexpected payment equal to one month of your after-tax income. Would you have sufficient financial resources, such as access to credit, savings, or loans from relatives or friends, to pay for the entire amount?*

- Yes
- No

Question 10 *Which of the following best characterizes your household?*

- Own my house/apartment without a mortgage/loan
- Own my house/apartment and have a fixed-rate mortgage/loan
- Own my house/apartment and have a variable-rate mortgage/loan
- Rent my house/apartment
- Other
- Don't know/ Prefer not to answer

Question 11 *Does your household have total financial investments (excluding housing) worth more than one month of combined household income?*

- Yes
- No
- Don't know/No opinion

Question 12 *Approximately how many times per month do you go to the gas station to buy gasoline or for other reasons?*

- Never
- Once per month
- Two to five times per month
- Six to ten times per month
- More than ten times per month

Question 13 *Who typically does the grocery shopping **in your household**?*

- I do all of the grocery shopping in the household
- I share the grocery shopping with others in the household

- Someone else does the grocery shopping in the household
- Don't know/No opinion

[IF Question 13==1:3]

Question 14 *How often do you, or others in your household, shop for groceries?*

- Every day
- Three or more times per week
- Twice per week
- Once per week
- Once every two weeks
- Once a month or less
- Don't know/No opinion

Question 15 *Who makes most financial decisions in your household?*

- I make most decisions
- Somebody else makes most decisions
- I make most decisions together with somebody else

As you may know, inflation is the percentage rise in overall prices in the economy. For example, if a price increases from 10 to 11, the rate of inflation is $(11-10)/10 = 10\%$. If a price decreases from 10 to 9, the rate of inflation is $(9-10)/10 = -10\%$.

We would like to ask you about the current rates of inflation in your country.

[INSTRUCTIONS] (If you do not think there was any inflation in the last 12 months, please enter a "0".)

Question 16 *Over the **last** 12 months, what do you believe the rate of inflation has been in your country?*

[OPEN END - FORCED NUMERIC, RANGE -100 to 1000, ALLOW FOR UP TO 2 DECIMAL POINTS]

Question 17 *Over the **next** 12 months, what do you expect the **lowest** possible rate of inflation could be in your country?*

[OPEN END - FORCED NUMERIC, RANGE -100 to 1000, ALLOW FOR UP TO 2 DECIMAL POINTS]

Question 18 *Over the **next** 12 months, what do you expect the **highest** possible rate of inflation could be in your country?*

[OPEN END - FORCED NUMERIC, RANGE -100 to 1000, ALLOW FOR UP TO 2 DECIMAL POINTS] [ENSURE VALUE IS GREATER THAN OR EQUAL TO ANSWER FROM QUESTION 17]

Question 19 *Over the **next** 12 months, what do you expect the **most likely** possible rate of inflation could be in your country?*

[OPEN END - FORCED NUMERIC, RANGE -100 to 1000, ALLOW FOR UP TO 2 DECIMAL POINTS] [ENSURE VALUE IS BETWEEN ANSWERS FROM QUESTION 17 AND QUESTION 18]

Question 20 *How certain are you that the actual rate of inflation in your country over the next 12 months will be within 2 percentage points higher or lower than [INSERT NUMERIC RESPONSE FROM QUESTION 19]?*

- Very certain
- Somewhat certain
- Somewhat uncertain
- Very uncertain

Question 21 *Thinking about your previous responses about the rate of inflation, were there any specific prices of goods that you had in mind when responding? Please select all that apply*

[RANDOMIZE; SELECT ALL THAT APPLY]

- Eggs/Produce
- Dairy/Milk
- Meat/Poultry
- Bread
- Rent
- Gasoline/Fuel/Petroleum
- Clothing
- Travel/Flight prices
- Other [ANCHOR] [OPEN END]
- Don't know/No opinion [ANCHOR]

Question 22 *Over the next 12 months, I expect the average nominal wage or salary of all workers in my country to ...*

[OPEN END – FORCED NUMERIC, RANGE -100 to 1000, ALLOW FOR UP TO 2 DECIMAL POINTS] [ALLOW ONLY ONE ANSWER]

- Go up by [] percent OR
- Go down by [] percent OR

- Stay about the same as today

Question 23 *Over the next 12 months, I expect the average house price in my country to ...*

[OPEN END – FORCED NUMERIC, RANGE -1000 to 1000, ALLOW FOR UP TO 2 DECIMAL POINTS] [ALLOW ONLY ONE ANSWER]

- Go up by [] percentage points OR
- Go down by [] percentage points OR
- Stay about the same as today

Question 24 *Over the next 12 months, I expect the level of interest rates in my country to ...*

[OPEN END – FORCED NUMERIC, RANGE -1000 to 1000, ALLOW FOR UP TO 2 DECIMAL POINTS] [ALLOW ONLY ONE ANSWER]

- Go up by [] percentage points OR
- Go down by [] percentage points OR
- Stay about the same as today

Question 25 *How important are the following sources for you to get information about inflation?*

[MATRIX; RANDOMIZE]

- Newspapers
- Television
- Social Media
- Government reports
- Personal shopping experience
- Energy bills
- Friends, family, and co-workers
- Your work activity, like customers, suppliers, competitors
- Please select “Don’t know/No opinion” [TERMINATE IF != 5]

1. Very important
2. Somewhat important
3. Not very important
4. Not at all important
5. Don’t know/No opinion

Question 26 *Considering the general economic situation, do you think now is the right time for people in your country to make major purchases such as furniture or electrical appliances?*

- Yes, now is the right time
- It is neither the right time nor the wrong time
- No, now is not the right time
- Don't know/No opinion

For the next couple of questions, we would like you to think about some hypothetical situations. Even if you don't have savings or income of your own, please answer these questions assuming you do.

Question 27 *In your opinion, is this statement true or false:*

If the money in your savings account grows at an annual rate of 5%, then you will be able to buy more with this money in the future than you are able to buy today regardless of the rate of inflation.

- True
- False
- Don't know/No opinion

Question 28 *How willing are you to take risks regarding financial matters?*

- Very willing
- Somewhat willing
- Not too willing
- Not at all willing
- Don't know/No opinion

Question 29 *Suppose you had saved some money and are considering what to do with it. Which, if any, of these actions would you take with this money?*

[SELECT ALL THAT APPLY RANDOMIZE]

- I would put the money in my bank account
- I would keep the money in cash
- I would invest the money in a fixed-return savings account even if it requires me to not withdraw the money for two to three years
- I would use the money to purchase an apartment/house
- I would use the money to invest in the stock market

- I would use the money to buy cryptocurrencies or other alternative investments
- I would give the money to my family or send it overseas as a remittance
- I would use the money to buy gold or foreign currency as a store of value
- None of the above [ANCHOR]

Question 30 *How well, if at all, do the following sentences describe you as a person?*

[MATRIX, RANDOMIZE]

- I am good at math
- I make impulsive purchasing decisions
- I ask for advice before making decisions about my finances
- I tend to postpone tasks even if I know it would be better to do them right away
- I am very impatient

1. Very well
2. Somewhat well
3. Not too well
4. Not at all well
5. Don't know/No opinion

Question 31 *How much trust do you have in each of the following institutions to act in the best interest of households?*

[MATRIX, RANDOMIZE]

- Your country's central bank
- Your country's government [ASK OF ALL COUNTRIES EXCEPT EGYPT, SAUDI ARABIA, UAE, MOROCCO, CHINA, RUSSIA, ZIMBABWE]
- Your country's banks and financial institutions
- Your country's business owners and corporations
- Your country's stock market
- International Financial Institutions (International Monetary Fund, World Bank, etc.)
- International banks and private financial institutions (Goldman Sachs, etc.)

1. A lot
2. Some

3. Not too much
4. Not at all
5. Don't know/No opinion

Question 32 *Is the party that best aligns with your political views currently part of the government in your home country? [ASK OF ALL COUNTRIES EXCEPT EGYPT, SAUDI ARABIA, UAE, MOROCCO, CHINA, RUSSIA, ZIMBABWE]*

- Yes
- No
- Prefer not to answer